

LATTICES OF THEORIES

In this chapter we use the duality theorem between quotients and subtoposes established in Chapter 3 to interpret many concepts of elementary topos theory which apply to the lattice of subtoposes of a given topos in the context of geometric theories.

Notions that we analyse in detail include the coHeyting algebra structure on the lattice of subtoposes of a given topos, open, closed, quasi-closed subtoposes, the Booleanization and DeMorganization of a topos, the dense-closed factorization of a geometric inclusion, skeletal inclusions, the surjection-inclusion factorization of a geometric morphism, atoms in the lattice of subtoposes of a given topos, and subtoposes with enough points.

We also obtain an explicit description of the Heyting operation between Grothendieck topologies on a given category and of the Grothendieck topology generated by a given collection of sieves, as well as a number of results about the problem of ‘relativizing’ a local operator with respect to a given subtopos.

4.1 The lattice operations on Grothendieck topologies and quotients

In this section we study the structure of the lattice of subtoposes of a given Grothendieck topos. It is well known that this lattice, endowed with the obvious order relation given by the inclusion of subtoposes, is a coHeyting algebra (see, for instance, section A4.5 of [55]). Our aim is to describe this structure in terms of Grothendieck topologies and then of theories in view of Theorem 3.2.5.

Given a Heyting algebra H and an element $a \in H$, the set $\uparrow(a)$ of elements $h \in H$ such that $h \geq a$ is closed under the operations of conjunction, disjunction and Heyting implication and is therefore a Heyting algebra with respect to these operations. Indeed, the assertion about the conjunction and disjunction is obvious, while the fact that $b \Rightarrow c$ is in $\uparrow(a)$ if b and c are follows from the inequality $c \leq (b \Rightarrow c)$.

This remark allows us to restrict our attention to the case of subtoposes of a presheaf topos in order to describe the operations of union, intersection and coHeyting implication on a pair of subtoposes of a given Grothendieck topos; indeed, the union (resp. intersection, coHeyting implication) of two subtoposes of $\mathbf{Sh}(\mathcal{C}, J)$ is the same as their union (resp. intersection, coHeyting implication) in the coHeyting algebra of subtoposes of $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$, since the order relation in the former lattice is the restriction of the order relation in the latter.

4.1.1 The lattice operations on Grothendieck topologies

Let $\mathcal{E} = [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ be a presheaf topos, with subobject classifier Ω . Recall that $\Omega : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$ is defined by

$$\begin{aligned}\Omega(c) &= \{R \mid R \text{ is a sieve on } c\} \text{ (for any object } c \in \mathcal{C}\text{);} \\ \Omega(f) &= f^*(-) \text{ (for any arrow } f \text{ in } \mathcal{C}\text{),}\end{aligned}$$

where $f^*(-)$ denotes the operation of pullback of sieves in $\mathcal{C}$ along f .

We know from Theorem 1.3.36 that, given a small category $\mathcal{C}$, the Grothendieck topologies J on $\mathcal{C}$ correspond exactly to local operators on the topos $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$; this correspondence, to which we refer as $(*)$, sends a local operator $j : \Omega \rightarrow \Omega$ to the subobject $J \rightarrow \Omega$ which it classifies, that is to the Grothendieck topology J on $\mathcal{C}$ defined by setting $S \in J(c)$ if and only if $j(c)(S) = M_c$, and conversely a subobject $J \rightarrow \Omega$ to the arrow $j : \Omega \rightarrow \Omega$ classifying it.

Given a subobject $A \rightarrow \Omega$, its classifying arrow $\chi_A : \Omega \rightarrow \Omega$ is given by the formula

$$\chi_A(c)(S) = \{f : d \rightarrow c \mid f^*(S) \in A(d)\}$$

(cf. the description of the classifying arrow of a subobject after Theorem 1.1.12).

It will be convenient for our purposes to give an explicit description of the internal Heyting operations $\wedge, \vee, \Rightarrow : \Omega \rightarrow \Omega$ on our topos $\mathcal{E}$. To this end, we recall from Lemma A1.6.3 [55] the abstract definitions of these operations (cf. also the proof of Theorem 1.3.29).

The internal conjunction map $\wedge : \Omega \times \Omega \rightarrow \Omega$ is the classifying arrow of the subobject $< \top, \top > : 1 \rightarrow \Omega \times \Omega$, whence

$$\wedge(c)(S, T) = S \cap T$$

for any object $c \in \mathcal{C}$ and sieves S and T on c .

The internal disjunction map $\vee : \Omega \times \Omega \rightarrow \Omega$ is the classifying arrow of the union of the subobjects $\pi_1^*(\top)$ and $\pi_2^*(\top)$ where π_1 and π_2 are the two product projections $\Omega \times \Omega \rightarrow \Omega$, whence

$$\vee(c)(S, T) = \{f : d \rightarrow c \mid f^*(S) \cup f^*(T) = M_d\}$$

for any object $c \in \mathcal{C}$ and sieves S and T on c .

The internal implication map $\Rightarrow : \Omega \times \Omega \rightarrow \Omega$ is the classifying arrow of the equalizer $\Omega_1 \rightarrow \Omega \times \Omega$ of $\wedge$ and π_1 , whence

$$\Rightarrow(c)(S, T) = \{f : d \rightarrow c \mid f^*(S) \subseteq f^*(T)\}$$

for any object $c \in \mathcal{C}$ and sieves S and T on c .

It is immediate to see that the order relation between local operators on $\mathcal{E}$ given by the opposite of the natural order between subtoposes transfers via $(*)$ to the following order between Grothendieck topologies on $\mathcal{C}$: $J \leq J'$ if and only if $J(c) \subseteq J'(c)$ for every $c \in \mathcal{C}$, that is if and only if every J -covering sieve is

J' -covering. Hence, the relation $\leq$ defines a Heyting algebra structure on the collection of Grothendieck topologies on the category $\mathcal{C}$; in particular, for any Grothendieck topologies J and J' on $\mathcal{C}$, there exist a meet $J \wedge J'$, a join $J \vee J'$ and a Heyting implication $J \Rightarrow J'$. The bottom element of this lattice is the Grothendieck topology $\perp$ on $\mathcal{C}$ given by $\perp(c) = \{M_c\}$ for every $c \in \mathcal{C}$, while the top element is the topology $\top$ defined by $\top(c) = \{S \mid S \text{ sieve on } c\}$ for every $c \in \mathcal{C}$.

It is immediate to see that the meet $J \wedge J'$ of two Grothendieck topologies J and J' on $\mathcal{C}$ is given by the following formula: for any object c of $\mathcal{C}$ and any sieve S on c , $S \in (J \wedge J')(c)$ if and only if $S \in J(c)$ and $S \in J'(c)$; indeed, the class of Grothendieck topologies is clearly closed under intersection. The join $J \vee J'$ is the smallest Grothendieck topology K such that $J \leq K$ and $J' \leq K$, so it is the Grothendieck topology generated by the collection of sieves which are either J -covering or J' -covering. In order to get a more explicit description of this operation, as well as of that of Heyting implication, we apply A. Joyal's theory as described in section A4.5 of [55]; this will lead in particular to an explicit description of the Grothendieck topology generated by a family of sieves which is stable under pullback.

First, let us explicitly describe, in terms of the category $\mathcal{C}$, the Galois connection from $\text{Sub}_{\mathcal{E}}(\Omega)$ to itself given by the mappings $D \rightarrow D^r$ and $D \rightarrow D^l$ described at p. 213 of [55].

Given a subobject $D \rightarrow \Omega$, the subobjects $D^r \rightarrow \Omega$ and $D^l \rightarrow \Omega$ are respectively defined to be

$$\forall_{\pi_2}((\pi_1^*(D) \Rightarrow \Theta) \rightarrow \Omega$$

and

$$\forall_{\pi_1}((\pi_2^*(D) \Rightarrow \Theta) \rightarrow \Omega,$$

where π_1 and π_2 are the two product projections $\Omega \times \Omega \rightarrow \Omega$, π_1^* and π_2^* are the pullback functors $\text{Sub}(\Omega) \rightarrow \text{Sub}(\Omega \times \Omega)$ respectively along π_1 and π_2 , $\Theta \rightarrow \Omega \times \Omega$ is the equalizer of $\pi_2, \Rightarrow: \Omega \times \Omega \rightarrow \Omega$, and $\forall_{\pi_1}, \forall_{\pi_2}$ are the right adjoints respectively to π_1^* and π_2^* .

From the above formulas, we get the following expression for Θ :

$$\Theta(c) = \{(S, T) \mid S \text{ and } T \text{ are sieves on } c \text{ such that for all } f: d \rightarrow c, \\ f^*(S) \subseteq f^*(T) \text{ implies } f \in T\},$$

for any object $c \in \mathcal{C}$.

Now, by using formula (7) at p. 146 of [63], we obtain

$$(\pi_1^*(D) \Rightarrow \Theta)(c) = \{(S, T) \mid S \text{ and } T \text{ are sieves on } c \text{ such that for all } f: d \rightarrow c, \\ [f^*(S) \in D(d) \text{ and } f^*(S) \subseteq f^*(T)] \text{ implies } f \in T\}.$$

By using formula (15) at p. 148 of [63], we get the following description of $\forall_{\pi_2}(A)$ for a subobject A of $\Omega \times \Omega$:

$$\forall_{\pi_2}(A)(c) = \{R \text{ sieve on } c \mid \text{for all } f : d \rightarrow c, \Omega \times f^*(R) \subseteq A\},$$

for any object $c \in \mathcal{C}$. If we apply this formula to the subobject $\pi_1^*(D) \Rightarrow \Theta$ calculated above we obtain

$$D^r(c) = \{T \text{ sieve on } c \mid \text{for any arrows } e \xrightarrow{h} d \xrightarrow{g} c \text{ and sieve } S \text{ on } d, \\ [h^*(S) \in D(e) \text{ and } h^*(S) \subseteq h^*(g^*(T))] \text{ implies } h \in g^*(T)\}.$$

Similarly, we derive the following expression for D^l :

$$D^l(c) = \{S \text{ sieve on } c \mid \text{for any arrows } e \xrightarrow{h} d \xrightarrow{g} c \text{ and sieve } T \text{ on } d, \\ [h^*(T) \in D(e) \text{ and } h^*(g^*(S)) \subseteq h^*(T)] \text{ implies } h \in T\}.$$

Notice that the above formulas can alternatively be put in the following form:

$$D^r(c) = \{T \text{ sieve on } c \mid \text{for any arrow } d \xrightarrow{f} c \text{ and sieve } S \text{ on } d, \\ [S \in D(d) \text{ and } S \subseteq f^*(T)] \text{ implies } f \in T\};$$

$$D^l(c) = \{S \text{ sieve on } c \mid \text{for any arrow } d \xrightarrow{f} c \text{ and sieve } Z \text{ on } d, \\ [Z \in D(d) \text{ and } f^*(S) \subseteq Z] \text{ implies } Z = M_d\}.$$

For instance, the equivalence of the former expression for D^l with this latter formulation can be verified by taking $g = f$, $h = 1_d$ and $T = Z$ in one direction and $f = g \circ h$ and $Z = h^*(T)$ in the other direction.

From these expressions one immediately obtains the following formula:

$$(D^r)^l(c) = \{S \text{ sieve on } c \mid \text{for any arrow } d \xrightarrow{f} c \text{ and sieve } T \text{ on } d, \\ [(\text{for any arrow } e \xrightarrow{g} d \text{ and sieve } Z \text{ on } e \\ (Z \in D(e) \text{ and } Z \subseteq g^*(T)) \text{ implies } g \in T) \text{ and } (f^*(S) \subseteq T)] \\ \text{implies } T = M_d\}.$$

We recall from the proof of Corollary A4.5.13(i) [55] that the classifying arrow of $(D^r)^l$ is the smallest local operator on $\mathcal{E}$ for which all the monomorphisms in $\mathcal{E}$ whose classifying arrow factors through $D \rightarrow \Omega$ are dense. Let us now show that, via the identification $(*)$, this local operator corresponds precisely to the Grothendieck topology generated by D (i.e. the smallest Grothendieck topology J on $\mathcal{C}$ such that all the sieves in D , regarded here as a collection of sieves in $\mathcal{C}$, are J -covering). To this end, it suffices to recall that, given a local operator j on a topos $\mathcal{E}$, the c_j -dense monomorphisms are exactly the monomorphisms whose classifying arrow factors through the subobject classified by j ; notice that if $\mathcal{E} = [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ and j corresponds to a Grothendieck topology J on $\mathcal{C}$, this subobject is exactly J (regarded as a subobject of $\Omega_{[\mathcal{C}^{\text{op}}, \mathbf{Set}]}$). Now, all the sieves in D are J -covering if and only if $D \leq J$ as subobjects of Ω , so our claim immediately follows. So our formula for $(D^r)^l$ gives an explicit description of the Grothendieck topology generated by D , as recorded by the following result:

Proposition 4.1.1 *Let $\mathcal{C}$ be a small category and D a family of sieves in $\mathcal{C}$ which is stable under pullback. Then the Grothendieck topology G_D generated by D is equal to $(D^r)^l$ and given by*

$$G_D(c) = \{S \text{ sieve on } c \mid \text{for any arrow } d \xrightarrow{f} c \text{ and sieve } T \text{ on } d, \\ [(\text{for any arrow } e \xrightarrow{g} d \text{ and sieve } Z \text{ on } e \\ (Z \in D(e) \text{ and } Z \subseteq g^*(T)) \text{ implies } g \in T) \text{ and } (f^*(S) \subseteq T)] \\ \text{implies } T = M_d\}$$

for any object $c \in \mathcal{C}$. □

Remark 4.1.2 The Grothendieck topology on a small category $\mathcal{C}$ generated by a given family F of sieves in $\mathcal{C}$ is given by $G_{F_{\text{p.b.}}}$, where $F_{\text{p.b.}}$ is the collection of sieves in $\mathcal{C}$ which are pullbacks of sieves in F .

Starting from Corollary A4.5.13(i) [55], one can prove by similar means that our formula for D^l gives an explicit description of the largest Grothendieck topology J on $\mathcal{C}$ such that all the sieves in D are J -closed (one recalls that the classifying arrow of D^l is the largest local operator on $\mathcal{E}$ for which the monomorphisms in $\mathcal{E}$ whose classifying arrow factors through $D \rightarrow \Omega$ are closed and replaces, in the above discussion, the subobject J classifying c_j -dense monomorphisms with the subobject Ω_J classifying c_j -closed monomorphisms, namely the equalizer of the arrows $j, 1_\Omega : \Omega \rightarrow \Omega$).

As an application of the above formulas for D^l , let us derive an explicit formula for the Heyting operation on the collection of Grothendieck topologies on a given small category.

Example 4.5.14(f) [55] provides a description of the Heyting operation on the collection of local operators on a topos: given local operators j_1 and j_2 on a topos $\mathcal{E}$, $j_1 \Rightarrow j_2 = (J_1 \cap \Omega_{j_2})^l$ (where J_1 is the subobject of Ω corresponding to j_1 and Ω_{j_2} is the image of j_2). If $\mathcal{E} = [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ and j_1, j_2 correspond to Grothendieck topologies J_1, J_2 on $\mathcal{C}$ via $(*)$ then our (second) formula for D^l gives the following expression for $J_1 \Rightarrow J_2$:

$$(J_1 \Rightarrow J_2)(c) = \{S \text{ sieve on } c \mid \text{for any arrow } d \xrightarrow{f} c \text{ and sieve } Z \text{ on } d, \\ [Z \text{ is } J_1\text{-covering and } J_2\text{-closed and } f^*(S) \subseteq Z] \\ \text{implies } Z = M_d\}.$$

In particular, the pseudocomplement $\neg J$ of a Grothendieck topology J on $\mathcal{C}$ is given by the following formula:

$$\neg J(c) = \{S \text{ sieve on } c \mid \text{for any arrow } d \xrightarrow{f} c \text{ and sieve } Z \text{ on } d, \\ [Z \text{ is } J\text{-covering and } f^*(S) \subseteq Z] \text{ implies } Z = M_d\}.$$

Let us now prove directly that, given a category $\mathcal{C}$ and a collection D of sieves in $\mathcal{C}$ which is closed under pullback, the above formula for D^l always defines a Grothendieck topology on $\mathcal{C}$ and $(D^r)^l$ is the Grothendieck topology on $\mathcal{C}$

generated by D . This will ensure that our results also hold for a general, not necessarily small, category $\mathcal{C}$.

It is clear that the above formula for D^l yields a collection of sieves satisfying the maximality and stability axioms for Grothendieck topologies, so it remains to verify that it satisfies the transitivity axiom. Let R and S be sieves on $c \in \mathcal{C}$ such that $S \in D^l(c)$ and for each $s : a \rightarrow c$ in S , $s^*(R) \in D^l(a)$; we want to prove that $R \in D^l(c)$, i.e. that for any arrow $f : d \rightarrow c$ and sieve Z on d , ($Z \in D(d)$ and $f^*(R) \subseteq Z$) implies $Z = M_d$. Now, for any $h \in f^*(S)$, $h^*(f^*(R)) \subseteq h^*(Z)$ and hence $h \in Z$ since $(f \circ h)^*(R) \in D^l(\text{dom}(h))$. So $f^*(S) \subseteq Z$, whence $Z = M_d$ since $S \in D^l(c)$.

Let us now show that $(D^r)^l$ is the Grothendieck topology on $\mathcal{C}$ generated by D ; since we already know that $(D^r)^l$ is a Grothendieck topology, this amounts to verifying that $D \leq (D^r)^l$ and that for any Grothendieck topology K on $\mathcal{C}$ which contains D , $(D^r)^l \leq K$. The fact that $D \leq (D^r)^l$ follows immediately from the above formula for $(D^r)^l$ (to see that any $S \in D(c)$ belongs to $(D^r)^l(c)$ take $g = 1_d$ and $Z = f^*(S)$ in it). To show that any Grothendieck topology K on $\mathcal{C}$ containing D is contained in $(D^r)^l$, suppose that S is a sieve in $(D^r)^l(c)$; then S is K -covering if and only if $\bar{S}^K = M_c$. Now, if we take $f = 1_c$ and $T = \bar{S}^K$ in the formula for $(D^r)^l$ we obtain that $f^*(S) \subseteq T$ and that for any arrow $e \xrightarrow{g} d$ and sieve Z on e , [$Z \in D(e)$ and $Z \subseteq g^*(T)$] implies that $g^*(T)$ is K -covering and hence maximal (it being K -closed); the formula thus implies that T is maximal, as required.

We can also verify directly that the above formula for $J_1 \Rightarrow J_2$ satisfies the property of the Heyting implication between J_1 and J_2 , i.e. that for any Grothendieck topology K on $\mathcal{C}$, $K \wedge J_1 \leq J_2$ if and only if $K \leq (J_1 \Rightarrow J_2)$. Indeed, $(J_1 \Rightarrow J_2) \wedge J_1 \leq J_2$ since for every $((J_1 \Rightarrow J_2) \wedge J_1)$ -covering sieve S , $S \subseteq \bar{S}^{J_2}$ and hence $\bar{S}^{J_2}$ is maximal, i.e. S is J_2 -covering; in the other direction, if $K \wedge J_1 \leq J_2$ then, for any K -covering sieve S , [Z is J_1 -covering and J_2 -closed, and $f^*(S) \subseteq Z$] implies that Z is $K \wedge J_1$ -covering and hence J_2 -covering and J_2 -closed, i.e. maximal. Incidentally, this argument yields the following alternative expression for $J_1 \Rightarrow J_2$:

$$(J_1 \Rightarrow J_2)(c) = \{S \text{ sieve on } c \mid \text{for any arrow } d \xrightarrow{f} c \text{ and sieve } Z \text{ on } d, \\ [Z \text{ is } J_1\text{-covering and } f^*(S) \subseteq Z] \text{ implies } Z \text{ is } J_2\text{-covering}\}$$

for any $c \in \mathcal{C}$.

4.1.2 The lattice operations on theories

By using Theorem 3.2.5 we can transfer the lattice operations on the collection of Grothendieck topologies on the geometric syntactic category $\mathcal{C}_{\mathbb{T}}$ of a geometric theory $\mathbb{T}$ which contain the syntactic topology $J_{\mathbb{T}}$ into the context of quotients of $\mathbb{T}$.

Let us denote by $\mathfrak{Th}_{\Sigma}^{\mathbb{T}}$ the collection of syntactic-equivalence classes of geometric theories over Σ which are quotients of $\mathbb{T}$. By definition of the duality of

Theorem 3.2.5, the order on $\mathfrak{H}_{\Sigma}^{\mathbb{T}}$ corresponding to the order $\leq$ between Grothendieck topologies on $\mathcal{C}_{\mathbb{T}}$ is the following: $\mathbb{T}' \leq \mathbb{T}''$ if and only if all the geometric sequents provable in $\mathbb{T}'$ (equivalently, all the geometric sequents in a given collection of axioms for $\mathbb{T}'$) are provable in $\mathbb{T}''$. So Theorem 3.2.5 gives the following result:

Theorem 4.1.3 *Let $\mathbb{T}$ be a geometric theory over a signature Σ . Then the collection $\mathfrak{H}_{\Sigma}^{\mathbb{T}}$ of (syntactic-equivalence classes of) geometric theories over Σ which are quotients of $\mathbb{T}$, endowed with the order defined by ' $\mathbb{T}' \leq \mathbb{T}''$ ' if and only if all the axioms of $\mathbb{T}'$ are provable in $\mathbb{T}''$ ', is a Heyting algebra.* $\square$

By taking $\mathbb{T}$ to be the empty (geometric) theory $\mathbb{O}_{\Sigma}$ over Σ , we obtain in particular that the collection $\mathfrak{H}_{\Sigma}^{\mathbb{O}_{\Sigma}}$ of all the (syntactic-equivalence classes of) geometric theories over Σ is a Heyting algebra.

Recalling the definition of the order in $\mathfrak{H}_{\Sigma}^{\mathbb{T}}$, we obtain the following descriptions of the lattice operations in $\mathfrak{H}_{\Sigma}^{\mathbb{T}}$:

- (i) The bottom element is $\mathbb{T}$.
- (ii) The top element is the *contradictory theory*, i.e. the theory axiomatized by $(\top \vdash_{\square} \perp)$ (in which all the geometric sequents over Σ are provable).
- (iii) The wedge $\mathbb{T}' \wedge \mathbb{T}''$ is the largest geometric theory over Σ which is contained in both $\mathbb{T}'$ and $\mathbb{T}''$, i.e. the collection of geometric sequents σ over Σ such that σ is provable in both $\mathbb{T}'$ and $\mathbb{T}''$.
- (iv) The join $\mathbb{T}' \vee \mathbb{T}''$ is the (syntactic-equivalence class of) the smallest geometric theory over Σ which contains both $\mathbb{T}'$ and $\mathbb{T}''$, i.e. the theory whose axioms are given by the union of the axioms of $\mathbb{T}'$ with those of $\mathbb{T}''$.
- (v) The Heyting implication $\mathbb{T}' \Rightarrow \mathbb{T}''$ is the (syntactic-equivalence class of the) largest geometric theory $\mathbb{S}$ over Σ such that $\mathbb{S} \wedge \mathbb{T}' \leq \mathbb{T}''$, i.e. such that every geometric sequent σ which is provable in both $\mathbb{S}$ and $\mathbb{T}'$ is provable in $\mathbb{T}''$; in particular, the pseudocomplement $\neg \mathbb{T}'$ is the (syntactic-equivalence class of the) largest geometric theory over Σ such that every geometric sequent σ which is provable in both $\mathbb{S}$ and $\mathbb{T}'$ is provable in $\mathbb{T}$.

We note that these operations are quite natural from a logical viewpoint; however it is by no means obvious from the perspective of geometric logic that there should exist a Heyting operation on the lattice of syntactic-equivalence classes of geometric theories over a given signature, while this follows as a formal consequence of Theorem 3.2.5. Another consequence of the theorem is the fact that the lattices $\mathfrak{H}_{\Sigma}^{\mathbb{T}}$ are complete; indeed, any intersection of Grothendieck topologies is a Grothendieck topology.

Figure 4.1 represents the isomorphic lattice structures present on the collection of geometric quotients of theories $\mathbb{T}$, $\mathbb{S}$, $\mathbb{R}$, etc. classified by a given topos $\mathcal{E}$.

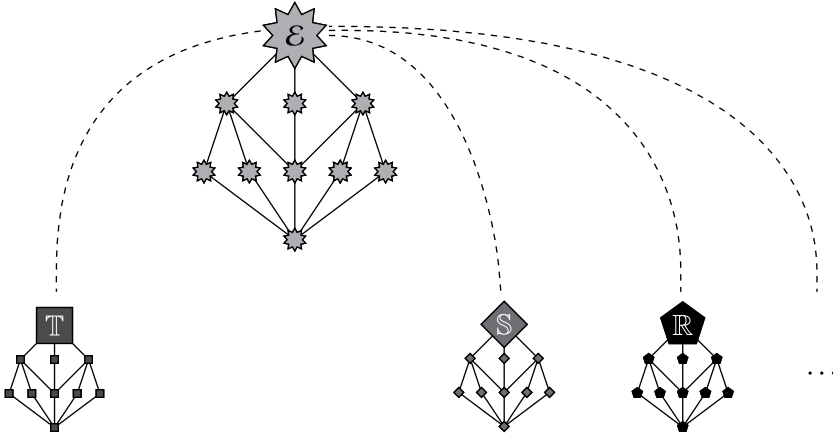


Fig. 4.1: Lattices of theories

Let us discuss, from the point of view of geometric logic, the fact that our lattice $\mathfrak{Th}_{\Sigma}^{\mathbb{T}}$ is distributive. This is a formal consequence of the fact that it is a Heyting algebra, which is ensured by Theorem 3.2.5, but it is instructive to establish this result directly by means of purely logical arguments. Our condition means that for any (syntactic-equivalence classes of) geometric theories $\mathbb{T}'$ and $\{\mathbb{T}_k \mid k \in K\}$ in $\mathfrak{Th}_{\Sigma}^{\mathbb{T}}$, $\mathbb{T}' \wedge (\bigvee_{k \in K} \mathbb{T}_k) = \bigvee_{k \in K} (\mathbb{T}' \wedge \mathbb{T}_k)$. Since the inequality $\geq$ is trivially satisfied, we are reduced to showing that for any geometric sequent σ over Σ , if σ is provable in $\mathbb{T}'$ and is derivable from axioms of the $\mathbb{T}_k$, then σ is derivable from axioms of the $\mathbb{T}' \wedge \mathbb{T}_k$. To this end, the following lemma is useful.

Lemma 4.1.4 *Let Σ be a first-order signature. If a geometric sequent $\sigma \equiv (\phi \vdash_{\vec{x}} \psi)$ over Σ is provable in the theory $\mathbb{S} = \{\tau \equiv (\phi_{\tau} \vdash \psi_{\tau}) \mid \tau \in \mathbb{S}\}$ using geometric logic then σ is provable in the theory $\mathbb{S}_{\sigma} = \{(\phi_{\tau} \wedge \phi \vdash \psi_{\tau} \vee \psi) \mid \tau \in \mathbb{S}\}$ using geometric logic.*

Proof Given a geometric sequent $\tau \equiv (\chi \vdash_{\vec{x}} \xi)$ over Σ , for a string of variables $\vec{x}'$ of the same kind as $\vec{x}$ we denote by $\mathcal{W}_{\vec{x}'}(\tau)$ the sequent $(\chi \wedge \phi[\vec{x}'/\vec{x}] \vdash \xi \vee \psi[\vec{x}'/\vec{x}])$. It is easy to check that for any instance of an inference rule of geometric logic, if we choose a string $\vec{x}'$ of variables that do not appear in any of the sequents involved in it then the image via $\mathcal{W}_{\vec{x}'}$ of the conclusion of the rule is derivable in geometric logic from the images via $\mathcal{W}_{\vec{x}'}$ of the premises of the rule. This clearly implies our thesis. $\square$

Now, our claim can be deduced as follows. If we have a derivation of a sequent $\sigma \equiv (\phi \vdash_{\vec{x}} \psi)$ over Σ which is provable in $\mathbb{T}'$ from sequents $\tau \equiv (\phi_{\tau} \vdash \psi_{\tau})$ each of which is an axiom of some $\mathbb{T}_k$ then by Lemma 4.1.4 we have a derivation

of σ from the sequents $(\phi \wedge \phi_\tau \vdash \psi \vee \psi_\tau)$, each of which is, on the one hand, provable in $\mathbb{T}'$ since it is derivable from σ , and, on the other hand, provable in $\mathbb{T}_k$ whenever τ lies in $\mathbb{T}_k$ since it is derivable from τ .

This is an illustration of the usefulness of Theorem 3.2.5 in providing insights into geometric logic; we shall discuss other applications of it below.

4.1.3 The Heyting implication in $\mathfrak{Th}_\Sigma^\mathbb{T}$

The purpose of this section is to give an explicit description of the Heyting operation on quotients of a given geometric theory $\mathbb{T}$. We will achieve this by interpreting the formula for the Heyting implication of Grothendieck topologies obtained in section 4.1.1 at the level of theories via the duality of Theorem 3.2.5.

Given two local operators j and j' on a topos $\mathcal{E}$, we write $j \leq j'$ to mean $\mathbf{sh}_{c_{j'}}(\mathcal{E}) \subseteq \mathbf{sh}_{c_j}(\mathcal{E})$ where c_j and $c_{j'}$ are the universal closure operations corresponding to j and j' (as in Theorem 1.3.32).

The following lemma will be useful for our purposes.

Lemma 4.1.5 *Let $\mathcal{E}$ be an elementary topos and j, j' two local operators on $\mathcal{E}$ with associated universal closure operations c_j and $c_{j'}$. Then $j \leq j'$ if and only if, for every subobject m in $\mathcal{E}$, $c_j(m) \leq c_{j'}(m)$ (equivalently, $c_{j'}(m) = c_{j'}(c_j(m))$).*

Proof Let L_j and $L_{j'}$ be the cartesian reflectors on $\mathcal{E}$ associated respectively with the local operators j and j' . Recall from section 1.3.4 that, given a monomorphism $m : A' \rightarrowtail A$ in $\mathcal{E}$, $c_j(m)$ is defined by the pullback square

$$\begin{array}{ccc} c_j(A') & \longrightarrow & L_j A' \\ \downarrow c_j(m) & & \downarrow L_j m \\ A & \xrightarrow{\eta_A} & L_j A, \end{array}$$

where η is the unit of the reflection induced by L_j . If $j \leq j'$ then $L_{j'}(L_j(m)) \cong L_{j'}(m)$ since $L_{j'}$ factors through L_j and both L_j and $L_{j'}$ are cartesian reflectors. If we apply the pullback-preserving functor $L_{j'}$ to the above pullback square we thus obtain that $L_{j'}(c_j(m)) \cong L_{j'}(L_j(m)) \cong L_{j'}(m)$; from this it immediately follows by definition of $c_{j'}$ in terms of $L_{j'}$ that $c_{j'}(m) = c_{j'}(c_j(m))$. In particular, $c_j(m) \leq c_{j'}(m)$.

The converse direction immediately follows from the fact that j is the classifying arrow of $c_j(\top)$ for each local operator j (cf. the proof of Theorem 1.3.32). $\square$

Remark 4.1.6 It follows at once from Lemma 4.1.5 that if $j \leq j'$ then, for any subobject m , if m is $c_{j'}$ -closed then m is c_j -closed.

We shall also need the following results.

Proposition 4.1.7 *Let $\mathcal{C}$ be a regular category, J a Grothendieck topology on $\mathcal{C}$ such that $J \supseteq J_{\mathcal{C}}^{\text{reg}}$ and $r : d \rightarrow c$ a cover in $\mathcal{C}$. Then*

- (i) For any sieve R on c , $R \in J(c)$ if and only if $r^*(R) \in J(d)$.
- (ii) For any sieve R on c generated by a monomorphism, R is J -closed if and only if $r^*(R)$ is J -closed.
- (iii) For any sieve R on c , R is J -closed if and only if, for any monomorphism $f : b \rightarrow c$, $f^*(R) \in J(b)$ implies $f \in R$.
- (iv) For any sieves R and T on c such that T is generated by a monomorphism, $r^*(R) \subseteq r^*(T)$ if and only if $R \subseteq T$.

Proof

- (i) This immediately follows from the stability and transitivity axioms for Grothendieck topologies.
- (ii) The ‘only if’ part is obvious; let us prove the ‘if’ part. Given an arrow $f : b \rightarrow c$ such that $f^*(R) \in J(b)$, we want to prove that $f \in R$. Consider the pullback

$$\begin{array}{ccc} a & \xrightarrow{h} & d \\ \downarrow g & & \downarrow r \\ b & \xrightarrow{f} & c \end{array}$$

in $\mathcal{C}$. From the commutativity of this square and the stability axiom for Grothendieck topologies it follows that $h^*(r^*(R)) \in J(a)$ and hence that $h \in r^*(R)$ i.e. $r \circ h \in R$. But $f \circ g = r \circ h$, so $f \circ g \in R$. Since g is a cover, R is generated by a monomorphism by our hypothesis and covers are orthogonal to monomorphisms (cf. Lemma A1.3.2 [55]), we conclude that $f \in R$, as required.

- (iii) The ‘only if’ part is obvious, so it remains to prove that if for any monomorphism $f : b \rightarrow c$, $f^*(R) \in J(b)$ implies $f \in R$, then R is J -closed. Let $g : e \rightarrow c$ be an arrow such that $g^*(R) \in J(e)$; we want to prove that $g \in R$. Denoting by $e \xrightarrow{g''} u \xrightarrow{g'} c$ the cover-mono factorization of g , we have by part (i) of the proposition that $g'^*(R) \in J(u)$; so $g' \in R$ by our hypothesis and hence $g \in R$, as required.
- (iv) The ‘if’ part is obvious, so it remains to prove that if $r^*(R) \subseteq r^*(T)$ then $R \subseteq T$. Given $f \in R$, consider the pullback

$$\begin{array}{ccc} a & \xrightarrow{h} & d \\ \downarrow g & & \downarrow r \\ b & \xrightarrow{f} & c \end{array}$$

in $\mathcal{C}$. Now, h belongs to $r^*(R)$ and hence to $r^*(T)$, so $f \circ g = r \circ h \in T$. But g is a cover and T is generated by a monomorphism so, since covers are orthogonal to monomorphisms (cf. Lemma A1.3.2 [55]), $f \in T$, as required.

□

Proposition 4.1.8 *Let $\mathbb{T}$ be a geometric theory over a signature Σ , $\mathbb{T}'$ a quotient of $\mathbb{T}$ and $\{\{\vec{x}_i.\phi_i\} \xrightarrow{[\theta_i]} \{\vec{y}.\psi\} \mid i \in I\}$ a set of generators for a sieve S in the syntactic category $\mathcal{C}_{\mathbb{T}}$ of $\mathbb{T}$. Then*

- (i) *S is $J_{\mathbb{T}'}^{\mathbb{T}}$ -covering if and only if the sequent $(\psi \vdash_{\vec{y}} \bigvee_{i \in I} (\exists \vec{x}_i) \theta_i)$ is provable in $\mathbb{T}'$.*
- (ii) *S is $J_{\mathbb{T}'}^{\mathbb{T}}$ -closed if and only if it is generated by a single monomorphism and for any geometric formula $\psi'(\vec{y})$ over Σ such that $(\psi' \vdash_{\vec{y}} \psi)$ is provable in $\mathbb{T}$, the sequent $(\psi' \vdash_{\vec{y}} \bigvee_{i \in I} (\exists \vec{x}_i) \theta_i)$ is provable in $\mathbb{T}'$ (if and) only if it is provable in $\mathbb{T}$.*

Proof

- (i) This is precisely equivalence (i) in Remark 3.2.8.
- (ii) This follows from Remark 4.1.6, Proposition 3.1.6(ii), Proposition 4.1.7 (iii) and part (i) of this proposition, by recalling the identification of subobjects of $\{\vec{y}.\psi\}$ in $\mathcal{C}_{\mathbb{T}}$ with $\mathbb{T}$ -provable equivalence classes of geometric formulae $\psi'(\vec{y})$ over Σ such that $(\psi' \vdash_{\vec{y}} \psi)$ is provable in $\mathbb{T}$ (cf. Lemma 1.4.2).

□

Now, let us suppose that $\mathcal{C}$ is the syntactic category $\mathcal{C}_{\mathbb{T}}$ of a geometric theory $\mathbb{T}$ and that J_1 and J_2 are respectively the associated $\mathbb{T}$ -topologies $J_{\mathbb{T}_1}^{\mathbb{T}}$ and $J_{\mathbb{T}_2}^{\mathbb{T}}$ of two quotients $\mathbb{T}_1$ and $\mathbb{T}_2$ of $\mathbb{T}$ (in the sense of section 3.2.1).

Recall from section 4.1.1 that

$$(J_1 \Rightarrow J_2)(c) = \{S \text{ sieve on } c \mid \text{for every arrow } d \xrightarrow{f} c \text{ and sieve } Z \text{ on } d, \\ [Z \text{ is } J_1\text{-covering and } f^*(S) \subseteq Z] \text{ implies } Z \text{ is } J_2\text{-covering}\}$$

for any $c \in \mathcal{C}$. In this formula, the quantification over all the sieves Z can be restricted to all the sieves Z of the form (t) for a monic arrow t . To show this, we notice that, by Proposition 3.1.6, $\overline{Z}^{J_{\mathbb{T}}}$ is generated by a monic arrow, and Z is J_1 -covering (resp. J_2 -covering) if and only if $\overline{Z}^{J_{\mathbb{T}}}$ is. Indeed, if Z is J_1 -covering (resp. J_2 -covering) then clearly $\overline{Z}^{J_{\mathbb{T}}}$ is J_1 -covering (resp. J_2 -covering) while the converse implication follows from the transitivity axiom for Grothendieck topologies in light of the fact that $J_{\mathbb{T}} \subseteq J_1$ (resp. $J_{\mathbb{T}} \subseteq J_2$).

We thus obtain the following simplified formula:

$$(J_1 \Rightarrow J_2)(c) = \{S \text{ sieve on } c \mid \text{for any arrow } d \xrightarrow{f} c \text{ and sieve } T = (t) \text{ on } d \text{ with } \\ t \text{ monic, } [T \text{ is } J_1\text{-covering and } f^*(S) \subseteq T] \text{ implies } \\ T \text{ is } J_2\text{-covering}\}$$

for any $c \in \mathcal{C}$.

Let us apply this formula to the syntactic category of a geometric theory $\mathbb{T}$. By Proposition 3.1.5(ii), $J_1 \Rightarrow J_2$ is generated over $J_{\mathbb{T}}$ by sieves generated by a single

monic arrow. This remark enables us to arrive at a simplified axiomatization of the Heyting implication $\mathbb{T}_1 \Rightarrow \mathbb{T}_2$, as follows.

Recall that, by Lemma 1.4.2, any subobject in $\mathcal{C}_{\mathbb{T}}$ is isomorphic to one of the form

$$\{\vec{y}' . \psi[\vec{y}'/\vec{y}]\} \xrightarrow{[\psi' \wedge \vec{y}' = \vec{y}]} \{\vec{y} . \psi\},$$

where $\psi(\vec{y})$ is a geometric formula such that the sequent $(\psi' \vdash_{\vec{y}} \psi)$ is provable in $\mathbb{T}$.

On the other hand, for any arrows

$$\{\vec{x} . \phi\} \xrightarrow{[\theta]} \{\vec{y} . \psi\}$$

and

$$\{\vec{y}' . \psi[\vec{y}'/\vec{y}]\} \xrightarrow{[\psi' \wedge \vec{y}' = \vec{y}]} \{\vec{y} . \psi\}$$

in $\mathcal{C}_{\mathbb{T}}$ with common codomain, we have the following pullback square in $\mathcal{C}_{\mathbb{T}}$:

$$\begin{array}{ccc} \{\vec{x} . (\exists \vec{y})(\theta[\vec{x}/\vec{x}] \wedge \psi')\} & \xrightarrow{[(\exists \vec{y})(\theta \wedge \psi' \wedge \vec{y}' = \vec{y}')] } & \{\vec{y}' . \psi[\vec{y}'/\vec{y}]\} \\ \downarrow [(\exists \vec{y})(\theta \wedge \psi' \wedge \vec{x} = \vec{x})] & & \downarrow [\psi' \wedge \vec{y}' = \vec{y}] \\ \{\vec{x} . \phi\} & \xrightarrow{[\theta]} & \{\vec{y} . \psi\}. \end{array}$$

By Lemma 1.4.2, the sieve generated by the arrow

$$[(\exists \vec{y})(\theta \wedge \psi' \wedge \vec{x} = \vec{x})] : \{\vec{x} . (\exists \vec{y})(\theta[\vec{x}/\vec{x}] \wedge \psi')\} \rightarrow \{\vec{x} . \phi\}$$

is contained in the sieve generated by an arrow

$$\{\vec{x}' . \chi[\vec{x}'/\vec{x}]\} \xrightarrow{[\chi \wedge \vec{x}' = \vec{x}]} \{\vec{x} . \phi\}$$

if and only if the sequent $((\exists \vec{y})(\theta(\vec{x}, \vec{y}) \wedge \psi'(\vec{y})) \vdash_{\vec{x}} \chi)$ is provable in $\mathbb{T}$.

In light of Proposition 4.1.8(i), we thus obtain the following result:

Theorem 4.1.9 *Let $\mathbb{T}$ be a geometric theory over a signature Σ and $\mathbb{T}_1, \mathbb{T}_2$ two quotients of $\mathbb{T}$. Then $\mathbb{T}_1 \Rightarrow \mathbb{T}_2$ is the theory obtained from $\mathbb{T}$ by adding all the geometric sequents $(\psi \vdash_{\vec{y}} \psi')$ over Σ with the property that $(\psi' \vdash_{\vec{y}} \psi)$ is provable in $\mathbb{T}$ and for any $\mathbb{T}$ -provably functional geometric formula $\theta(\vec{x}, \vec{y})$ from a geometric formula-in-context $\{\vec{x} . \phi\}$ to $\{\vec{y} . \psi\}$ and any geometric formula χ in the context $\vec{x}$ such that $(\chi \vdash_{\vec{x}} \phi)$ is provable in $\mathbb{T}$, the conjunction of the facts*

- (i) $(\phi \vdash_{\vec{x}} \chi)$ is provable in $\mathbb{T}_1$
- (ii) $((\exists \vec{y})(\theta(\vec{x}, \vec{y}) \wedge \psi'(\vec{y})) \vdash_{\vec{x}} \chi)$ is provable in $\mathbb{T}$

implies that $(\phi \vdash_{\vec{x}} \chi)$ is provable in $\mathbb{T}_2$. □

In particular, we obtain that the pseudocomplement of a quotient $\mathbb{T}'$ in $\mathfrak{Th}_\Sigma^\mathbb{T}$ is the theory $\neg\mathbb{T}' = (\mathbb{T}' \Rightarrow \mathbb{T})$ obtained from $\mathbb{T}$ by adding all the geometric sequents $(\psi \vdash_{\vec{y}} \psi')$ over Σ with the property that $(\psi' \vdash_{\vec{y}} \psi)$ is provable in $\mathbb{T}$ and for any $\mathbb{T}$ -provably functional geometric formula $\theta(\vec{x}, \vec{y})$ from a geometric formula-in-context $\{\vec{x}. \phi\}$ to $\{\vec{y}. \psi\}$ and any geometric formula χ in the context $\vec{x}$ such that $(\chi \vdash_{\vec{x}} \phi)$ is provable in $\mathbb{T}$, the conjunction of the facts

- (i) $(\phi \vdash_{\vec{x}} \chi)$ is provable in $\mathbb{T}'$
- (ii) $((\exists \vec{y})(\theta(\vec{x}, \vec{y}) \wedge \psi'(\vec{y})) \vdash_{\vec{x}} \chi)$ is provable in $\mathbb{T}$

implies that $(\phi \vdash_{\vec{x}} \chi)$ is provable in $\mathbb{T}$.

4.2 Transfer of notions from topos theory to logic

In this section we describe in detail the logical counterparts, via the duality of Theorem 3.2.5, of a number of important topos-theoretic notions which apply to the lattice of subtoposes of a given topos.

The following preliminary section, which concerns the problem of relativizing a local operator with respect to another one, will be instrumental in connection with calculations concerning open and quasi-closed local operators on a topos.

4.2.1 Relativization of local operators

Recall from Theorem 1.3.32 that for any topos $\mathcal{E}$ there is a bijection between universal closure operations on $\mathcal{E}$ and local operators on $\mathcal{E}$: this bijection sends a local operator $j : \mathcal{E} \rightarrow \mathcal{E}$ to the universal closure operation c_j (also denoted c_L , where L is the corresponding reflector on $\mathcal{E}$) defined, for each monomorphism $m : A' \rightarrowtail A$ in $\mathcal{E}$, by the pullback square

$$\begin{array}{ccc} c_L(A') & \longrightarrow & LA' \\ \downarrow & & \downarrow Lm \\ A & \xrightarrow{\eta_A^L} & LA, \end{array}$$

where L is the cartesian reflector on $\mathcal{E}$ corresponding to j and η_A^L is the unit of the reflection, and a universal closure operation c on $\mathcal{E}$ to the local operator $j_c : \Omega \rightarrow \Omega$ given by the classifying arrow of the subobject $c(1 \xrightarrow{\top} \top)$. Recall also that, given a local operator j on $\mathcal{E}$, the domain Ω_j of the equalizer $e_j : \Omega_j \rightarrowtail \Omega$ of the arrows $1_\Omega, j : \Omega \rightarrow \Omega$ is the subobject classifier of the topos $\mathbf{sh}_j(\mathcal{E})$ and the classifying arrow $\chi_m : A \rightarrow \Omega$ of a monomorphism m in $\mathcal{E}$ factors through e_j if and only if m is c_j -closed.

Given geometric inclusions $\mathcal{F} \xrightleftharpoons[L']{i'} \mathcal{F}$ and $\mathcal{F} \xrightleftharpoons[L]{i} \mathcal{E}$, let us denote by $j_{L'}$

and j_L the corresponding local operators respectively on $\mathcal{F}$ and on $\mathcal{E}$. Denoting by Ω the subobject classifier of $\mathcal{E}$, we define $e_L : \Omega_L \rightarrowtail \Omega$ to be the equalizer of $1_{\Omega_L}, j_L : \Omega_L \rightarrow \Omega$, $e_{L'} : (\Omega_L)_{L'} \rightarrowtail \Omega_L$ to be the equalizer of $1_{\Omega_L}, j_{L'} : \Omega_{L'} \rightarrow \Omega_L$ and $e_{L' \circ L} : \Omega_{L' \circ L} \rightarrowtail \Omega$ to be the equalizer of $1_\Omega, j_{L' \circ L} : \Omega \rightarrow \Omega$.

Lemma 4.2.1 *With the above notation, the composite arrow*

$$(\Omega_L)_{L'} \xrightarrow{e_{L'}} \Omega_L \xrightarrow{e_L} \Omega$$

and the arrow

$$\Omega_{L' \circ L} \xrightarrow{e_{L' \circ L}} \Omega$$

are isomorphic (as objects of $\mathcal{E}/\Omega$).

Proof Let us first prove that, given a subobject $m : A' \rightarrowtail A$ in $\mathcal{E}$ with classifying arrow $\chi_m : A \rightarrow \Omega$,

- (1) χ_m factors through $e_L \circ e_{L'}$ if and only if $(c_L(m) = m$ and $c_{L'}(Lm) = Lm)$;
- (2) χ_m factors through $e_{L' \circ L}$ if and only if $c_{L' \circ L}(m) = m$;
- (3) $(c_L(m) = m$ and $c_{L'}(Lm) = Lm)$ if and only if $c_{L' \circ L}(m) = m$.

(1) χ_m factors through $e_L \circ e_{L'}$ if and only if χ_m factors through e_L and the factorization χ_m^L of χ_m through e_L factors through $e_{L'}$; by definition of e_L , the first condition means precisely that $c_L(m) = m$, while the second, in view of the adjunction $\text{Hom}_{\mathbf{sh}_{j_L}(\mathcal{E})}(LA, \Omega_L) \cong \text{Hom}_{\mathcal{E}}(A, \Omega_L)$, is equivalent to the requirement that the subobject in $\mathbf{sh}_{j_L}(\mathcal{E})$ classified by the factorization $\overline{\chi_m^L} : LA \rightarrow \Omega_L$ of χ_m^L through $\eta_A^L : A \rightarrow LA$ be $c_{L'}$ -closed (by definition of $e_{L'}$).

Consider the diagram

$$\begin{array}{ccccc} A' & \xrightarrow{!} & 1 & \xrightarrow{!} & 1 \\ \downarrow m & & \downarrow \top_L & & \downarrow \top \\ A & \xrightarrow{\chi_m^L} & \Omega_L & \xrightarrow{e_L} & \Omega, \end{array}$$

where $\top_L$ is the factorization of $\top : 1 \rightarrow \Omega$ through e_L . The outer rectangle is the pullback square witnessing that χ_m classifies m , while the right-hand square is trivially a pullback (e_L being monic); the pullback lemma thus implies that the left-hand square is a pullback. Since L preserves pullbacks, the square

$$\begin{array}{ccc} LA' & \xrightarrow{!} & 1 \\ \downarrow Lm & & \downarrow \top_L \\ LA & \xrightarrow{\overline{\chi_m^L}} & \Omega_L \end{array}$$

is a pullback, i.e. $\overline{\chi_m^L}$ classifies the subobject Lm in $\mathbf{sh}_{j_L}(\mathcal{E})$. This concludes the proof of (1).

- (2) This is immediate by definition of $\Omega_{L' \circ L}$.

(3) By definition of $c_{L'}$ and $c_{L' \circ L}$, we have a rectangle

$$\begin{array}{ccccc} c_{L' \circ L}(A) & \longrightarrow & c_{L'}(LA') & \longrightarrow & L'(LA') \\ \downarrow c_{L' \circ L}(m) & & \downarrow c_{L'}(Lm) & & \downarrow L'(Lm) \\ A & \xrightarrow{\eta_A^L} & LA & \xrightarrow{\eta_{LA}^{L'}} & L'(LA) \end{array}$$

in which both squares are pullbacks; this follows as an immediate consequence of the pullback lemma, since $\eta_A^{L' \circ L} = \eta_{LA}^{L'} \circ \eta_A^L$. Notice in particular that if A is an L -sheaf then $c_{L' \circ L}(m) = c_{L'}(Lm)$.

Suppose that $c_L(m) = m$ and $c_{L'}(Lm) = Lm$. The fact that $c_{L'}(Lm) = Lm$ implies, by definition of $c_L(m)$ and the fact that the left-hand square above is a pullback, that $c_{L' \circ L}(m) = c_L(m)$; hence, $c_L(m) = m$ implies $c_{L' \circ L}(m) = m$, as required. Conversely, suppose that $c_{L' \circ L}(m) = m$. By applying the pullback-preserving functor L to the left-hand square above, we obtain that $Lm = c_{L'}(Lm)$; but then, by definition of $c_L(m)$, we have $c_{L' \circ L}(m) = c_L(m)$ and hence $c_L(m) = m$. This concludes the proof of (3).

Now, (1), (2) and (3) together imply that for any subobject m in $\mathcal{E}$, χ_m factors through $e_L \circ e_{L'}$ if and only if it factors through $e_{L' \circ L}$. Our thesis thus follows from the Yoneda lemma. $\square$

The following definition will be central for the results in this section:

Definition 4.2.2 Given a topos $\mathcal{E}$, two local operators j and k on $\mathcal{E}$ and a local operator $k' : \Omega_j \rightarrow \Omega_j$ in $\mathbf{sh}_j(\mathcal{E})$, we say that k *relativizes to k' at j* (or that k' is the *relativization of k at j*) if the diagram

$$\begin{array}{ccc} \Omega_j & \xrightarrow{k'} & \Omega_j \\ \downarrow e_j & & \downarrow e_j \\ \Omega & \xrightarrow{k} & \Omega \end{array}$$

in $\mathcal{E}$ commutes.

Notice that, in the above definition, since e_j is monic, there can be at most one relativization of k at j .

The fundamental properties of relativizations are expressed by the following result:

Theorem 4.2.3 *Let k' be the relativization of k at j as above. Then*

- (i) $\mathbf{sh}_{k'}(\mathbf{sh}_j(\mathcal{E})) = \mathbf{sh}_{k \vee j}(\mathcal{E})$ (where $k \vee j$ is the join of k and j in the lattice of local operators on $\mathcal{E}$).
- (ii) For any subobject m in $\mathbf{sh}_j(\mathcal{E})$, $c_{k'}(m) = c_k(m)$.
- (iii) If $k \geq j$ then, for any subobject m in $\mathcal{E}$, $c_{k'}(L_j m) = c_k(m)$.

Proof

- (i) Let s be the local operator on $\mathcal{E}$ corresponding to $\mathbf{sh}_{k'}(\mathbf{sh}_j(\mathcal{E}))$, regarded as a subtopos of $\mathcal{E}$ via the composite geometric inclusion

$$\mathbf{sh}_{k'}(\mathbf{sh}_j(\mathcal{E})) \hookrightarrow \mathbf{sh}_j(\mathcal{E}) \hookrightarrow \mathcal{E}.$$

We have to prove that $e_s : \Omega_s \rightarrow \Omega$ is isomorphic to $e_{k \vee j} : \Omega_{k \vee j} \rightarrow \Omega$. By the Yoneda lemma, it is equivalent to prove that for any subobject $m : A' \rightarrow A$, χ_m factors through e_s if and only if it factors through $e_{k \vee j}$. Now, by Lemma 4.2.1, χ_m factors through e_s if and only if m is c_j -closed and Lm is $c_{k'}$ -closed, where L is the cartesian reflector corresponding to j , while, by Example A4.5.13 [55], χ_m factors through $e_{k \vee j}$ if and only if m is both c_j -closed and c_k -closed. So we have to prove that, given a c_j -closed subobject $m : A' \rightarrow A$, Lm is $c_{k'}$ -closed if and only if m is c_k -closed. Consider the commutative diagram

$$\begin{array}{ccccc} A & \xrightarrow{\chi_m} & \Omega & \xrightarrow{k} & \Omega \\ \eta_A^L \downarrow & \searrow \chi_m^L & \uparrow e_j & & \uparrow e_j \\ LA & \xrightarrow{\overline{\chi_m^L}} & \Omega_j & \xrightarrow{k'} & \Omega_j, \end{array}$$

where the notation is that of the proof of Lemma 4.2.1.

From the proof of Lemma 4.2.1 we know that $\overline{\chi_m^L}$ is the characteristic map in $\mathbf{sh}_j(\mathcal{E})$ of the subobject Lm . By definition of Ω_k , χ_m factors through e_k (i.e. m is c_k -closed) if and only if $k \circ \chi_m = \chi_m$, while, by definition of $\Omega_{k'}^{\mathbf{sh}_j(\mathcal{E})}$, $\overline{\chi_m^L}$ factors through $\Omega_{k'}^{\mathbf{sh}_j(\mathcal{E})} \rightarrow \Omega_j$ (i.e. Lm is $c_{k'}$ -closed) if and only if $k' \circ \overline{\chi_m^L} = \chi_m^L$. Now, since e_j is monic and η_A^L is the unit of the reflection corresponding to j , $k' \circ \overline{\chi_m^L} = \chi_m^L$ if and only if $e_j \circ k' \circ \overline{\chi_m^L} \circ \eta_A^L = e_j \circ \chi_m^L \circ \eta_A^L$. But by the commutativity of the above diagram, this is precisely equivalent to $k \circ \chi_m = \chi_m$.

- (ii) The condition $k \circ e_j = e_j \circ k' : \Omega_j \rightarrow \Omega$ is equivalent to the assertion that the subobjects classified by the maps $k \circ e_j$ and $e_j \circ k'$ are equal. Now, since k classifies $c_k(\top)$, $k \circ e_j$ classifies $e_j^*(c_k(\top)) = c_k(e_j^*(\top)) = c_k(\top_j)$, where $\top_j$ is the factorization of $\top$ through e_j , while $e_j \circ k'$ is easily seen to classify $c_{k'}(\top_j)$. So the condition amounts to requiring that $c_{k'}(\top_j) = c_k(\top_j)$. But every subobject in $\mathbf{sh}_j(\mathcal{E})$ is a pullback (both in $\mathbf{sh}_j(\mathcal{E})$ and in $\mathcal{E}$) of $\top_j$; hence for any subobject m in $\mathbf{sh}_j(\mathcal{E})$, $c_{k'}(m) = c_k(m)$, as required.
- (iii) By (ii), it suffices to prove that if m is a subobject in $\mathcal{E}$ then $c_k(L_j m) = c_k(m)$; this immediately follows from the definition of $c_k(-)$ as the pullback of $L_k(-)$ along the unit of the reflector L_k and the fact that if $k \geq j$ then $L_k(m) \cong L_k(L_j(m))$.

□

Let us now consider some instances of relativizations.

Proposition 4.2.4 *With the notation of the proof of Lemma 4.2.1, $j_{L'} : \Omega_L \rightarrow \Omega_L$ is the relativization of $j_{L' \circ L}$ at j_L ; that is, the square*

$$\begin{array}{ccc} \Omega_{j_L} & \xrightarrow{j_{L'}} & \Omega_{j_L} \\ \downarrow e_{j_L} & & \downarrow e_{j_L} \\ \Omega & \xrightarrow{j_{L' \circ L}} & \Omega \end{array}$$

commutes.

Proof Let us prove that the composites $e_{j_L} \circ j_{L'}$ and $j_{L' \circ L} \circ e_{j_L}$ classify the same subobject of $\mathbf{sh}_{j_L}(\mathcal{E})$, namely $c_{L'}(\top_L)$.

Consider the diagram

$$\begin{array}{ccccc} c_{L'}(1) & \xrightarrow{!} & 1 & \xrightarrow{!} & 1 \\ \downarrow c_{L'}(\top_L) & & \downarrow \top_L & & \downarrow \top \\ \Omega_L & \xrightarrow{j_{L'}} & \Omega_L & \xrightarrow{e_{j_L}} & \Omega. \end{array}$$

Since both squares in it are pullbacks it follows from the pullback lemma that $e_{j_L} \circ j_{L'}$ classifies $c_{L'}(\top_L)$. On the other hand, if $j_{L' \circ L}$ classifies $c_{L' \circ L}(\top)$ then $j_{L' \circ L} \circ e_{j_L}$ classifies $e_{j_L}^*(c_{L' \circ L}(\top)) = c_{L' \circ L}(e_{j_L}^*(\top)) = c_{L' \circ L}(\top_L)$; but $c_{L' \circ L}(\top_L) = c_{L'}(\top_L)$ (cf. the proof of Lemma 4.2.1), which implies our thesis. $\square$

Remarks 4.2.5 (a) All the relativizations arising as in Proposition 4.2.4 have the property that $k \geq j$. We shall see below instances of relativizations not enjoying this property. For the moment, we note that if k' is the relativization at j of two local operators k_1 and k_2 then $k_1 \vee j = k_2 \vee j$. This follows at once from Theorem 4.2.3 by recalling the identification between subcategories of sheaves on a topos and local operators on it.

(b) Given k and j local operators on a topos $\mathcal{E}$, there exists a relativization of k at j if and only if $j \circ k \circ e_j = k \circ e_j$ (equivalently, $c_k(\top_j)$ being classified by $k \circ e_j$, if and only if $c_k(\top_j)$ is c_j -closed); in particular, if $k \geq j$ then k relativizes at j . Conversely, given a local operator k' on $\mathbf{sh}_j(\mathcal{E})$, there always exists a local operator k on $\mathcal{E}$ such that k relativizes to k' at j . Take, for instance, k to be the local operator on $\mathcal{E}$ corresponding to the composite of the canonical geometric inclusions $\mathbf{sh}_{k'}(\mathbf{sh}_j(\mathcal{E})) \hookrightarrow \mathbf{sh}_j(\mathcal{E})$ and $\mathbf{sh}_j(\mathcal{E}) \hookrightarrow \mathcal{E}$; then, by Proposition 4.2.4 and Remark 4.2.5(a), k relativizes to k' at j .

Proposition 4.2.6 *Under the hypotheses of Theorem 4.2.3, if k relativizes to k' at j then $k \vee j$ relativizes to k' at j .*

Proof The condition $(k \vee j) \circ e_j = e_j \circ k'$ is equivalent to the assertion that the arrows $(k \vee j) \circ e_j$ and $e_j \circ k'$ classify the same subobject, i.e. that $c_{k \vee j}(\top_j) =$

$c_k(\top_j)$. Now, since $k \leq k \vee j$, $c_{k \vee j}(\top_j) \geq c_k(\top_j)$. To show that $c_{k \vee j}(\top_j) \leq c_k(\top_j)$ it is enough to prove, by the characterization of the closure of a subobject as the smallest closed subobject containing it, that $c_k(\top_j)$ is $c_{k \vee j}$ -closed. We have observed in the proof of Theorem 4.2.3 that the $c_{k \vee j}$ -closed subobjects are exactly those which are both c_j -closed and c_k -closed; our thesis thus follows at once from Remark 4.2.5(b).

Alternatively, our thesis follows as a consequence of Theorem 4.2.3(i) and Proposition 4.2.4. $\square$

Let us now review the notions of open and quasi-closed local operator on a topos (cf. also section 4.2.2 below) and show that they behave naturally with respect to relativizations.

Recall from section A4.5 of [55] that the *open local operator* $o^\mathcal{E}(U)$ associated with a subterminal object U of a topos $\mathcal{E}$ is given by the composite arrow

$$\Omega \cong 1 \times \Omega \xrightarrow{u \times 1} \Omega \times \Omega \xrightarrow{\Rightarrow} \Omega,$$

where $u : 1 \rightarrow \Omega$ is the classifying arrow of the subobject $U \rightarrowtail 1$, while the *quasi-closed local operator* $qc^\mathcal{E}(U)$ associated with U is the composite arrow

$$\Omega \cong \Omega \times 1 \xrightarrow{1 \times \langle u, u \rangle} \Omega \times \Omega \times \Omega \xrightarrow{\Rightarrow \times 1} \Omega \times \Omega \xrightarrow{\Rightarrow} \Omega.$$

Given a local operator j on a topos $\mathcal{E}$ and a subterminal object U of $\mathbf{sh}_j(\mathcal{E})$, since the inclusion $\mathbf{sh}_j(\mathcal{E}) \hookrightarrow \mathcal{E}$ preserves monomorphisms, we can regard U as a subterminal object of $\mathcal{E}$ as well. Then we have the following result:

Proposition 4.2.7 *Let $\mathcal{E}$ be a topos and j a local operator on $\mathcal{E}$. Given a subterminal object U of $\mathbf{sh}_j(\mathcal{E})$, the open (resp. quasi-closed) local operator $o^{\mathbf{sh}_j(\mathcal{E})}(U)$ (resp. $qc^{\mathbf{sh}_j(\mathcal{E})}(U)$) in $\mathbf{sh}_j(\mathcal{E})$ associated with U is the relativization at j of the open (resp. quasi-closed) local operator $o^\mathcal{E}(U)$ (resp. $qc^\mathcal{E}(U)$) in $\mathcal{E}$ associated with U (regarded as a subterminal object of $\mathcal{E}$).*

Proof From the description of the internal Heyting operations $\wedge_\mathcal{E}, \vee_\mathcal{E}, \Rightarrow_\mathcal{E} : \Omega \rightarrow \Omega$ on $\mathcal{E}$ given in the proof of Theorem 1.3.29, it follows that the diagrams

$$\begin{array}{ccc} \Omega_j \times \Omega_j & \xrightarrow{\wedge_{\mathbf{sh}_j(\mathcal{E})}} & \Omega_j \\ \downarrow e_j \times e_j & & \downarrow e_j \\ \Omega \times \Omega & \xrightarrow{\wedge_\mathcal{E}} & \Omega \end{array} \quad \text{and} \quad \begin{array}{ccc} \Omega_j \times \Omega_j & \xrightarrow{\Rightarrow_{\mathbf{sh}_j(\mathcal{E})}} & \Omega_j \\ \downarrow e_j \times e_j & & \downarrow e_j \\ \Omega \times \Omega & \xrightarrow{\Rightarrow_\mathcal{E}} & \Omega \end{array}$$

commute. Let us first show that the left-hand square commutes. The arrow $\wedge_\mathcal{E} : \Omega \times \Omega \rightarrow \Omega$ is the classifying arrow of $\langle \top, \top \rangle : 1 \rightarrow \Omega \times \Omega$ and $\wedge_{\mathbf{sh}_j(\mathcal{E})} : \Omega_j \times \Omega_j \rightarrow \Omega_j$ is the classifying arrow of $\langle \top_j, \top_j \rangle : 1 \rightarrow \Omega_j \times \Omega_j$, that is of the factorization of $\langle \top, \top \rangle$ through $e_j \times e_j$. To prove that the composites

$e_j \circ \wedge_{\mathbf{sh}_j(\mathcal{E})}$ and $\wedge_{\mathcal{E}} \circ (e_j \times e_j)$ classify the same subobject of $\mathbf{sh}_j(\mathcal{E})$, namely $\langle \top_j, \top_j \rangle: 1 \rightarrow \Omega_j \times \Omega_j$, consider the following diagram:

$$\begin{array}{ccccc} 1 & \xrightarrow{\quad ! \quad} & 1 & \xrightarrow{\quad ! \quad} & 1 \\ \downarrow \langle \top_j, \top_j \rangle & & \downarrow \top_j & & \downarrow \top \\ \Omega_j \times \Omega_j & \xrightarrow{\quad \wedge_{\mathbf{sh}_j(\mathcal{E})} \quad} & \Omega_j & \xrightarrow{\quad e_j \quad} & \Omega. \end{array}$$

Since both squares in it are pullbacks, it follows from the pullback lemma that $e_j \circ \wedge_{\mathbf{sh}_j(\mathcal{E})}$ classifies $\langle \top_j, \top_j \rangle$. On the other hand, if $\wedge_{\mathcal{E}}$ classifies $\langle \top, \top \rangle$ then $\wedge_{\mathcal{E}} \circ e_j$ classifies $e_j^*(\langle \top, \top \rangle) = \langle \top_j, \top_j \rangle$. This proves that the square for $\wedge$ commutes.

Let us now show that the square for $\Rightarrow$ commutes. We observe that $\Omega \times \Omega \xrightarrow{\Rightarrow_{\mathcal{E}}} \Omega$ is the classifying arrow of $r: E \rightarrow \Omega \times \Omega$ and $\Omega_j \times \Omega_j \xrightarrow{\Rightarrow_{\mathbf{sh}_j(\mathcal{E})}} \Omega_j$ is the classifying arrow of $r_j: E_j \rightarrow \Omega_j \times \Omega_j$ where r and r_j are respectively the equalizers of $\wedge_{\mathcal{E}}, \pi_1^{\mathcal{E}}: \Omega \times \Omega \rightarrow \Omega$ and of $\wedge_{\mathbf{sh}_j(\mathcal{E})}, \pi_1^{\mathbf{sh}_j(\mathcal{E})}: \Omega_j \times \Omega_j \rightarrow \Omega_j$.

It is easy to see, by invoking the commutativity of the square for $\wedge$, that the pullback of r along $e_j \times e_j$ is an equalizer for $\wedge_{\mathbf{sh}_j(\mathcal{E})}, \pi_1^{\mathbf{sh}_j(\mathcal{E})}: \Omega_j \times \Omega_j \rightarrow \Omega_j$ and hence that it is isomorphic to r_j ; from this our claim follows immediately.

Now, by definition of open and quasi-closed local operators, the commutativity of the diagram for $\Rightarrow$ immediately implies our thesis, since if U is a subterminal object of $\mathbf{sh}_j(\mathcal{E})$ then the classifying arrow of $U \rightarrow 1$ in $\mathbf{sh}_j(\mathcal{E})$ is the factorization of its classifying arrow in $\mathcal{E}$ through $e_j: \Omega_j \rightarrow \Omega$. $\square$

Definition 4.2.8 A local operator j on a topos $\mathcal{E}$ is said to be *dense* if $j \leq \neg$.

A subtopos of a topos $\mathcal{E}$ is said to be *dense* if the corresponding local operator on $\mathcal{E}$ is dense.

As an application of Theorem 4.2.3 and Proposition 4.2.7, we recover the following well-known result:

Corollary 4.2.9 Let $\mathcal{E}$ be topos and j be a dense local operator on $\mathcal{E}$. Then $\mathbf{sh}_{\neg}(\mathbf{sh}_j(\mathcal{E})) = \mathbf{sh}_{\neg}(\mathcal{E})$.

Proof For any topos $\mathcal{E}$, $qc^{\mathcal{E}}(0_{\mathcal{E}}) = \neg\neg_{\mathcal{E}}$ (cf. p. 220 of [55]). Our thesis thus follows from Theorem 4.2.3 and Proposition 4.2.7 by invoking the fact (remarked at p. 219 of [55]) that for a dense local operator j on $\mathcal{E}$, the inclusion $\mathbf{sh}_j(\mathcal{E}) \hookrightarrow \mathcal{E}$ preserves the initial object. $\square$

Given an elementary topos $\mathcal{E}$, let us denote by $\mathbf{Lop}(\mathcal{E})$ the collection of local operators on $\mathcal{E}$, endowed with the Heyting algebra structure given by the canonical order between them. For any local operator j on $\mathcal{E}$, there is a bijection between the collection of local operators k in $\mathcal{E}$ such that $k \geq j$ and the collection of local operators on $\mathbf{sh}_j(\mathcal{E})$. Indeed, if $k \geq j$ then the geometric inclusion $\mathbf{sh}_k(\mathcal{E}) \hookrightarrow \mathcal{E}$ factors (uniquely up to isomorphism) through $\mathbf{sh}_j(\mathcal{E}) \hookrightarrow \mathcal{E}$ and hence it corresponds to a unique local operator k_j on $\mathbf{sh}_j(\mathcal{E})$ such that $\mathbf{sh}_{k_j}(\mathbf{sh}_j(\mathcal{E})) = \mathbf{sh}_k(\mathcal{E})$,

while conversely, given a local operator s on $\mathbf{sh}_j(\mathcal{E})$, the geometric inclusion given by the composite $\mathbf{sh}_s(\mathbf{sh}_j(\mathcal{E})) \hookrightarrow \mathbf{sh}_j(\mathcal{E}) \hookrightarrow \mathcal{E}$ corresponds to a unique local operator s^j on $\mathcal{E}$ such that $\mathbf{sh}_s(\mathbf{sh}_j(\mathcal{E})) = \mathbf{sh}_{s^j}(\mathcal{E})$. It is clear that these two correspondences are inverse to each other. Moreover, since the order between local operators on a topos corresponds to the reverse inclusion between the corresponding subcategories of sheaves, these bijections are also order-preserving (where the order between local operators k on $\mathcal{E}$ such that $k \geq j$ is the restriction of the order in $\mathbf{Lop}(\mathcal{E})$ and the order between local operators on $\mathbf{sh}_j(\mathcal{E})$ is the canonical one on $\mathbf{Lop}(\mathbf{sh}_j(\mathcal{E}))$). Recall that, given a Heyting algebra H and an element $a \in H$, $\uparrow(a) = \{b \in H \mid b \geq a\}$ is a Heyting algebra which is closed under the operations of conjunction, disjunction and Heyting implication in H and hence the map $a \vee (-) : H \rightarrow \uparrow(a)$ is a Heyting algebra homomorphism. So the bijections $(-)_j$ and $(-)^j$ are isomorphisms of Heyting algebras between the subalgebra $\uparrow(j)$ of $\mathbf{Lop}(\mathcal{E})$ and the algebra $\mathbf{Lop}(\mathbf{sh}_j(\mathcal{E}))$, whence the map $(j \vee (-))_j : \mathbf{Lop}(\mathcal{E}) \rightarrow \mathbf{Lop}(\mathbf{sh}_j(\mathcal{E}))$ is a Heyting algebra homomorphism.

4.2.2 Open, closed, quasi-closed subtoposes

Open, closed and quasi-closed subtoposes are particularly important classes of subtoposes, built from a given topos and a subterminal object of it. In this section we analyse the logical counterparts of these concepts in geometric logic, that is we describe the quotients of a given geometric theory corresponding via the duality of Theorem 3.2.5 to such subtoposes of its classifying topos.

4.2.2.1 Open subtoposes

Recall from section A4.5 of [55] that an open subtopos of a topos $\mathcal{E}$ is a geometric inclusion of the form $\mathcal{E}/U \hookrightarrow \mathcal{E}$ for a subterminal object U of $\mathcal{E}$. The associated universal closure operation sends a subobject $A' \rightrightarrows A$ to the implication $(A \times U) \Rightarrow A'$ in the Heyting algebra $\mathbf{Sub}_{\mathcal{E}}(A)$. If $L_U : \mathcal{E} \rightarrow \mathcal{E}$ is the corresponding cartesian reflector, we thus have that a monomorphism $A' \rightrightarrows A$ is L_U -dense if and only if $(A \times U) \leq A'$ in $\mathbf{Sub}_{\mathcal{E}}(A)$. So $A \times U$ is the smallest L_U -dense subobject of A , and hence L_U is the smallest local operator on $\mathcal{E}$ for which the monomorphism $U \rightrightarrows 1$ is dense (cf. the discussion preceding Lemma A4.5.10 [55]). From Proposition A4.3.11 [55] we thus deduce that a geometric morphism $f : \mathcal{F} \rightarrow \mathcal{E}$ factors through the inclusion $\mathcal{E}/U \hookrightarrow \mathcal{E}$ if and only if $f^*(U) = 1$.

Let $\mathcal{E}$ be the classifying topos $\mathbf{Set}[\mathbb{T}] \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$ of a geometric theory $\mathbb{T}$ over a signature Σ . We shall now describe the quotient of $\mathbb{T}$ corresponding via Theorem 3.2.5 to an open subtopos $\mathcal{E}/U \hookrightarrow \mathcal{E}$ of $\mathcal{E}$. Recall that the geometric syntactic category $\mathcal{C}_{\mathbb{T}}$ of $\mathbb{T}$ embeds into its ∞ -pretopos completion $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$ via the Yoneda embedding $y : \mathcal{C}_{\mathbb{T}} \hookrightarrow \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$, and under this identification all the subobjects in $\mathbf{Set}[\mathbb{T}]$ of an object in $\mathcal{C}_{\mathbb{T}}$ lie again in $\mathcal{C}_{\mathbb{T}}$. Since the terminal object of $\mathcal{E}$ can be identified with $\{\square, \top\}$ and the subobjects of a given object $\{\vec{x}, \psi\}$ of $\mathcal{C}_{\mathbb{T}}$ can be identified with the geometric formulae $\phi(\vec{x})$ which $\mathbb{T}$ -provably imply $\psi(\vec{x})$ (cf. Lemma 1.4.2), we conclude that the subterminal object U of 1 in $\mathcal{E}$ corresponds to a unique (up to $\mathbb{T}$ -provable equivalence) geometric sentence (i.e.

geometric formula with no free variables) ϕ over Σ . Alternatively, $\{\square \cdot \phi\}$ arises as the domain of the subobject of $\{\square \cdot \top\}$ given by the union of all the images of the morphisms from objects in $\{c \in \mathcal{C}_{\mathbb{T}} \mid U(c) = 1_{\mathbf{Set}} = \{*\}\}$ to the terminal object of $\mathcal{C}_{\mathbb{T}}$ (cf. Proposition 3.1.5).

Recall that Diaconescu's equivalence

$$\mathbf{Geom}(\mathcal{F}, \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})) \simeq \mathbf{Flat}_{J_{\mathbb{T}}}(\mathcal{C}_{\mathbb{T}}, \mathcal{F})$$

sends a geometric morphism $f : \mathcal{F} \rightarrow \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$ to the functor $f^* \circ y : \mathcal{C}_{\mathbb{T}} \rightarrow \mathcal{F}$ (where $y : \mathcal{C}_{\mathbb{T}} \rightarrow \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$ is the Yoneda embedding), while the equivalence

$$\mathbb{T}\text{-mod}(\mathcal{F}) \simeq \mathbf{Flat}_{J_{\mathbb{T}}}(\mathcal{C}_{\mathbb{T}}, \mathcal{F})$$

sends each model $M \in \mathbb{T}\text{-mod}(\mathcal{F})$ to the functor $F_M : \mathcal{C}_{\mathbb{T}} \rightarrow \mathcal{F}$ assigning to a formula $\{\vec{x} \cdot \phi\}$ its interpretation $[[\vec{x} \cdot \phi]]_M$ in M . Thus, via the composite equivalence

$$\mathbf{Geom}(\mathcal{F}, \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})) \simeq \mathbb{T}\text{-mod}(\mathcal{F}),$$

the geometric morphisms $\mathcal{F} \rightarrow \mathcal{E} = \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$ which factor through $\mathcal{E}/U \hookrightarrow \mathcal{E}$ correspond to the $\mathbb{T}$ -models M such that $[[\phi]]_M = 1$, i.e. such that ϕ is satisfied in M . It follows that the quotient $\mathbb{T}_{\phi}$ of $\mathbb{T}$ obtained by adding to $\mathbb{T}$ the axiom $(\top \vdash_{\square} \phi)$ is classified by the topos $\mathcal{E}/U$ and corresponds to it via the duality of Theorem 3.2.5.

Given a topos $\mathbf{Sh}(\mathcal{C}, J)$ and a subterminal object U of it, the subtopos $\mathbf{Sh}(\mathcal{C}, J)/U \hookrightarrow \mathbf{Sh}(\mathcal{C}, J)$ corresponds to a unique Grothendieck topology J_U^{open} on $\mathcal{C}$ such that $J_U^{\text{open}} \supseteq J$; in particular, we have $\mathbf{Sh}(\mathcal{C}, J)/U \simeq \mathbf{Sh}(\mathcal{C}, J_U^{\text{open}})$. By Theorem 4.2.3 and Proposition 4.2.7, $J_U^{\text{open}} = J \vee J_{o(U)}$, where $J_{o(U)}$ is the Grothendieck topology on $\mathcal{C}$ corresponding to the open local operator $o(U)$ on $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$ associated with U . Now, $o(U)$ is by definition given by the composite arrow

$$\Omega \cong 1 \times \Omega \xrightarrow{u \times 1} \Omega \times \Omega \xrightarrow{\Rightarrow} \Omega,$$

where Ω is the subobject classifier of $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$ and $u : 1 \rightarrow \Omega$ is the classifying arrow of the subobject U . As a subterminal object of $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$, U can be identified with the full subcategory $\mathcal{C}_U$ of $\mathcal{C}$ on the objects c such that $U(c) = \{*\}$. So $u(c)(*) = \{f : d \rightarrow c \mid d \in \mathcal{C}_U\}$ for any object $c \in \mathcal{C}$. Let us set, for any $c \in \mathcal{C}$, $Z(c) = u(c)(*)$. Then, by applying the general formula for $\Rightarrow$ recalled at the beginning of section 4.1.1, we obtain that $o(U)$ sends a sieve R on an object $c \in \mathcal{C}$ to $\{g : e \rightarrow c \mid g^*(Z(c)) \subseteq g^*(R)\}$. Hence $J_{o(U)}$ is given by

$$R \in J_{o(U)}(c) \text{ if and only if } R \supseteq Z(c)$$

for any $c \in \mathcal{C}$. In particular, by property (b) in Definition 3.1.3, $J_{o(U)}$ is generated by the sieves $Z(c)$, as c varies in $\mathcal{C}$. Notice in passing that for any arrow $f : d \rightarrow c$ in $\mathcal{C}$, $f^*(Z(c)) = Z(d)$.

Let us apply this discussion to the syntactic representation

$$\mathbf{Set}[\mathbb{T}] \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$$

of the classifying topos $\mathbf{Set}[\mathbb{T}]$ of a geometric theory $\mathbb{T}$ over a signature Σ . Identifying the subterminal U of $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$ with (the $\mathbb{T}$ -provable equivalence class of) a geometric sentence ϕ as above, we can describe the subcategory $\mathcal{C}_{\phi}$ corresponding to it as the full subcategory of $\mathcal{C}_{\mathbb{T}}$ on the objects $\{\vec{x}. \psi\}$ such that there exists (exactly) one morphism $\{\vec{x}. \psi\} \rightarrow \{\square. \phi\}$ in $\mathcal{C}_{\mathbb{T}}$. By recalling the definition of morphisms in the syntactic category $\mathcal{C}_{\mathbb{T}}$, one then immediately obtains the following characterization for the objects of $\mathcal{C}_{\phi}$: $\{\vec{x}. \psi\} \in \mathcal{C}_{\phi}$ if and only if the sequent $(\psi \vdash_{\vec{x}} \phi)$ is provable in $\mathbb{T}$. By definition of $\mathcal{C}_{\phi}$, the sieve $Z(\{\square. \top\})$ is generated by the morphism $\{\square. \phi\} \rightarrow \{\square. \top\}$. As the topology J_U^{open} is generated over $J_{\mathbb{T}}$ by the sieves $Z(c)$, and $Z(c)$ is the pullback of $Z(\{\square. \top\})$ along the unique arrow $c \rightarrow \{\square. \top\}$ for any $c \in \mathcal{C}_{\mathbb{T}}$, the quotient of $\mathbb{T}$ corresponding to the subtopos $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})/U \hookrightarrow \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$ is axiomatized over $\mathbb{T}$ by the sequent $(\top \vdash_{\square} \phi)$ (cf. Remark 3.2.9).

4.2.2.2 Closed subtoposes

Recall from section A4.5 of [55] that, given an elementary topos $\mathcal{E}$ and a subterminal object U of $\mathcal{E}$, the *closed local operator* $c^{\mathcal{E}}(U)$ associated with U is the composite arrow

$$\Omega \cong 1 \times \Omega \xrightarrow{u \times 1} \Omega \times \Omega \xrightarrow{\vee} \Omega,$$

where $u : 1 \rightarrow \Omega$ is the classifying arrow of the subobject U . Unlike open and quasi-closed local operators, a closed local operator on $\mathcal{E}$ associated with a subterminal object U lying in a subtopos $\mathbf{sh}_j(\mathcal{E})$ does not relativize in general to the closed local operator on $\mathbf{sh}_j(\mathcal{E})$ associated with U ; however, if $\mathcal{E}$ is the topos $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$ we can easily find a local operator on $\mathcal{E}$ which relativizes to $c^{\mathbf{Sh}(\mathcal{C}, J)}(U)$, as follows. The local operator $c^{\mathbf{Sh}(\mathcal{C}, J)}(U)$ sends any J -closed sieve R on $c \in \mathcal{C}$ to the $(J$ -closed) sieve $\{f : d \rightarrow c \mid f^*(Z(c)) \cup f^*(R) \in J(d)\}$. This naturally leads us to consider the arrow $\Omega_{[\mathcal{C}^{\text{op}}, \mathbf{Set}]} \rightarrow \Omega_{[\mathcal{C}^{\text{op}}, \mathbf{Set}]}$ in $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$ sending a sieve R on $c \in \mathcal{C}$ to the sieve $\{f : d \rightarrow c \mid f^*(Z(c)) \cup f^*(R) \in J(d)\}$. It is easily checked that this arrow is a local operator on $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$ which relativizes to $c^{\mathbf{Sh}(\mathcal{C}, J)}(U)$. Indeed, this arrow corresponds to the Grothendieck topology J_U^{closed} on $\mathcal{C}$ given by

$$R \in J_U^{\text{closed}}(c) \text{ if and only if } Z(c) \cup R \in J(c)$$

for any $c \in \mathcal{C}$. Since $J_U^{\text{closed}} \supseteq J$, J_U^{closed} is, by Theorem 4.2.3, the Grothendieck topology on $\mathcal{C}$ corresponding to the subtopos of $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$ given by the composite of the closed subtopos of $\mathbf{Sh}(\mathcal{C}, J)$ associated with U with the canonical geometric inclusion $\mathbf{Sh}(\mathcal{C}, J) \hookrightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$.

Let us now give a description of the theory $\mathbb{T}_{\phi}^{\text{closed}}$ over Σ corresponding via Theorem 3.2.5 to the closed subtopos

$$\mathbf{sh}_{\mathcal{C}^{\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})}(U)}(\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})) \hookrightarrow \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$$

of the classifying topos of $\mathbb{T}$ associated with the subterminal object U of the topos $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$, identified as above with a geometric sentence ϕ over Σ . Since $Z(c)$ is the pullback of $Z(\{\square \cdot \top\})$ along the unique arrow $c \rightarrow \{\square \cdot \top\}$ for any $c \in \mathcal{C}_{\mathbb{T}}$, Theorem 3.2.5 and Proposition 3.2.9 allow us to identify $\mathbb{T}_{\phi}^{\text{closed}}$ with the quotient of $\mathbb{T}$ obtained by adding all the the geometric sequents

$$(\psi \vdash_{\bar{y}} \psi')$$

such that the sequents $(\psi' \vdash_{\bar{y}} \psi)$ and $(\psi \vdash_{\bar{y}} \psi' \vee (\phi \wedge \psi))$ are provable in $\mathbb{T}$. A minimal axiomatization for $\mathbb{T}_{\phi}^{\text{closed}}$ over $\mathbb{T}$ is given by the unique axiom $(\phi \vdash_{\square} \perp)$.

4.2.2.3 Quasi-closed subtoposes

If $\mathcal{E}$ is the topos $\mathbf{Sh}(\mathcal{C}, J)$ and U is a subterminal object of $\mathbf{Sh}(\mathcal{C}, J)$ then $qc^{\mathcal{E}}(U)$ corresponds to a unique Grothendieck topology J_U^{qc} on $\mathcal{C}$ such that $J_U^{\text{qc}} \supseteq J$. By Theorem 4.2.3 and Proposition 4.2.7, this topology is $J \vee J_{qc(U)}$, where $J_{qc(U)}$ is the Grothendieck topology on $\mathcal{C}$ corresponding to the quasi-closed local operator $qc^{[\mathcal{C}^{\text{op}}, \mathbf{Set}]}(U)$ on $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$ associated with U (regarded here as a subterminal object of $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$).

As above, let us identify U with the full subcategory $\mathcal{C}_U$ of $\mathcal{C}$ on the objects c such that $U(c) = \{*\}$ and set, for any $c \in \mathcal{C}$, $Z(c) = u(c) = \{f : d \rightarrow c \mid d \in \mathcal{C}_U\}$. In the case $\mathcal{E} = [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ the local operator $qc^{\mathcal{E}}(U)$ is easily seen to send a sieve R on $c \in \mathcal{C}$ to the sieve $\Rightarrow(c)(\{f : d \rightarrow c \mid f^*(R) \subseteq f^*(Z(c))\}, Z(c))$. Hence $J_{qc(U)}$ is given by

$$R \in J_{qc(U)}(c) \text{ iff for any } f : d \rightarrow c, (f^*(R) \subseteq Z(d) \text{ implies } f \in Z(c))$$

for any $c \in \mathcal{C}$.

In order to specialize the above expression to the syntactic site of a geometric theory, we observe that, if $\mathcal{C}$ is a geometric category and J contains the geometric topology J^{geom} on $\mathcal{C}$ then the condition on the right-hand side of the equivalence is satisfied by $f : d \rightarrow c$ if and only if it is satisfied by the image $f' : d' \rightarrow c$ of f in $\mathcal{C}$. Indeed, since $\mathcal{C}_U$ is a J -ideal and every cover generates a J -covering sieve, $f \in Z(c)$ if and only if $f' \in Z(c)$. Let us now prove that for any $f : d \rightarrow c$, $f'^*(R) \subseteq Z(d')$ if and only if $f^*(R) \subseteq Z(d)$.

Since the $Z(c)$ are stable under pullback, $f'^*(R) \subseteq Z(d')$ implies $f^*(R) \subseteq Z(d)$. Conversely, let $r : d \twoheadrightarrow d'$ be the factorization of f through f' ; given $g \in f'^*(R)$, consider the pullback

$$\begin{array}{ccc} e & \xrightarrow{g'} & d \\ \downarrow r' & & \downarrow r \\ e' & \xrightarrow{g} & d' \end{array}$$

in $\mathcal{C}$. Clearly, since R is a sieve, $g' \in f^*(R) \subseteq Z(d)$ whence $e \in \mathcal{C}_U$; but r' is a cover, which implies that $e' \in \mathcal{C}_U$ and hence that $g \in Z(d')$, as required.

This remark enables us to achieve a simplified description of the theory $\mathbb{T}_\phi^{\text{qc}}$ over Σ corresponding via Theorem 3.2.5 to the quasi-closed subtopos $qc^{\mathbf{Set}[\mathbb{T}]}(U)$ of the classifying topos $\mathbf{Set}[\mathbb{T}] \simeq \mathbf{Sh}(\mathcal{C}_\mathbb{T}, J_\mathbb{T})$ of $\mathbb{T}$, where ϕ is the geometric sentence over Σ corresponding to a subterminal object U of $\mathbf{Set}[\mathbb{T}]$. Indeed, by recalling the identification between $\mathbb{T}$ -provable equivalence classes of geometric formulae $\psi'(\vec{y})$ such that $(\psi' \vdash_{\vec{y}} \psi)$ is provable in $\mathbb{T}$ and subobjects of $\{\vec{y}. \psi\}$ in $\mathcal{C}_\mathbb{T}$ given by Lemma 1.4.2, we get, by Theorem 3.2.5, Proposition 3.1.5(ii) and the syntactic characterization of $\mathcal{C}_\phi$ given above, the following axiomatization for $\mathbb{T}_\phi^{\text{qc}}$: $\mathbb{T}_\phi^{\text{qc}}$ is obtained from $\mathbb{T}$ by adding all the geometric sequents

$$(\psi \vdash_{\vec{y}} \psi')$$

such that $(\psi' \vdash_{\vec{y}} \psi)$ is provable in $\mathbb{T}$ and for any geometric formula $\chi(\vec{y})$ over Σ such that $(\chi \vdash_{\vec{y}} \psi)$ is provable in $\mathbb{T}$, if $(\chi \wedge \psi' \vdash_{\vec{y}} \phi)$ is provable in $\mathbb{T}$ then $(\chi \vdash_{\vec{y}} \phi)$ is provable in $\mathbb{T}$.

Remark 4.2.10 If ϕ is $\perp$ then, in view of Remark 3.2.9(b), $\mathbb{T}_\phi^{\text{qc}}$ specializes to the Booleanization of $\mathbb{T}$ described in section 4.2.3.

4.2.2.4 Open and closed quotients

Let $\mathbb{T}$ be a geometric theory over a signature Σ . Given an elementary topos $\mathcal{E}$, it is well known that the open and closed subtoposes associated with a given subterminal object are complementary to each other in $\mathbf{Lop}(\mathcal{E})$. From this we deduce, by Theorem 3.2.5, that the open and closed quotients $\mathbb{T}_\phi$ and $\mathbb{T}_\phi^{\text{closed}}$ of $\mathbb{T}$ corresponding to a given geometric sentence ϕ are complementary to each other in $\mathfrak{H}_\Sigma^\mathbb{T}$; in fact, this can also be proved directly by means of purely logical arguments. We also know from the theory of elementary toposes that if U and V are complemented subterminals of a topos $\mathcal{E}$ then $o(U) = c(V)$; this implies, by Theorem 3.2.5, that if ϕ and ψ are two geometric sentences such that $(\top \vdash_{\square} \phi \vee \psi)$ and $(\phi \wedge \psi \vdash_{\square} \perp)$ then $\mathbb{T}_\phi = \mathbb{T}_\psi^{\text{closed}}$; this as well is not hard to prove by using logical arguments.

4.2.3 The Booleanization and DeMorganization of a geometric theory

Recall that the *Booleanization* of a topos $\mathcal{E}$ is the subtopos $\mathbf{sh}_{\neg, \neg}(\mathcal{E}) \hookrightarrow \mathcal{E}$. This subtopos was characterized in [13] (cf. Theorem 2.4 therein) as the smallest (in fact, unique) Boolean dense subtopos of $\mathcal{E}$. The Booleanization of a topos $\mathcal{E}$ also coincides with the quasi-closed subtopos $\mathbf{sh}_{qc^{\mathcal{E}}(0_{\mathcal{E}})}(\mathcal{E}) \hookrightarrow \mathcal{E}$ (in the sense of section 4.2.2.3, cf. also Remark 4.2.10).

The notion of Booleanization is a topos-theoretic invariant admitting natural site characterizations; in particular, we have the following

Proposition 4.2.11

- (i) For any small category $\mathcal{C}$, the Booleanization of the topos $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$ is isomorphic to the canonical geometric inclusion $\mathbf{Sh}(\mathcal{C}, D) \hookrightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$, where D is the dense topology on $\mathcal{C}$, that is the Grothendieck topology

on $\mathcal{C}$ whose covering sieves are exactly the stably non-empty ones (cf. Corollary VI.5 [63]); in particular, if $\mathcal{C}$ satisfies the right Ore condition then the Booleanization of $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$ coincides with the canonical geometric inclusion

$$\mathbf{Sh}(\mathcal{C}, J_{\text{at}}) \hookrightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}],$$

where J_{at} is the atomic topology on $\mathcal{C}$.

- (ii) For any geometric theory $\mathbb{T}$, the Booleanization of its classifying topos can be identified with the subtopos corresponding via the duality of Theorem 3.2.5 to the quotient $\mathbb{T}_b$ of $\mathbb{T}$ obtained from $\mathbb{T}$ by adding all the sequents of the form

$$(\top \vdash_{\vec{y}} \psi)$$

for a geometric formula-in-context $\psi(\vec{y})$ over the signature of $\mathbb{T}$ which is $\mathbb{T}$ -stably consistent, i.e. such that for any geometric formula $\chi(\vec{y})$ in the same context, if $(\chi \vdash_{\vec{y}} \perp)$ is not provable in $\mathbb{T}$ then $(\chi \wedge \psi \vdash_{\vec{y}} \perp)$ is not provable in $\mathbb{T}$ (cf. Theorem 5.7 [13] or Remark 4.2.10).

We call $\mathbb{T}_b$ the Booleanization of the theory $\mathbb{T}$.

A number of concrete examples of Booleanizations of theories arising in different fields of mathematics is provided in section 10.4 below.

Recall that a topos $\mathcal{E}$ satisfies De Morgan's law if and only if, in every subobject lattice in the topos, the identity $\neg m \vee \neg \neg m = 1$ holds. In [13] (cf. Theorem 2.3 therein) it is shown that every topos $\mathcal{E}$ admits a largest dense subtopos satisfying De Morgan's law; this topos is called the *DeMorganization* of $\mathcal{E}$.

In view of the duality of Theorem 3.2.5, it is natural to wonder whether the logical counterpart of this construction can be explicitly described in syntactic terms. A uniform axiomatization of the *DeMorganization* $\mathbb{T}_m$ of a geometric theory $\mathbb{T}$ (i.e. of the quotient of $\mathbb{T}$ corresponding to the DeMorganization of its classifying topos) was provided in [13] (cf. Theorem 5.6 therein) and reads as follows: $\mathbb{T}_m$ is obtained from $\mathbb{T}$ by adding all the geometric sequents of the form

$$(\top \vdash_{\vec{x}} \bigvee_{\psi(\vec{x}) \in I_{(\phi(\vec{x}), \phi'(\vec{x}))}} \psi)$$

for any two geometric formulae $\phi(\vec{x})$ and $\phi'(\vec{x})$ over the signature of $\mathbb{T}$ such that $(\phi \wedge \phi' \vdash_{\vec{x}} \perp)$ is provable in $\mathbb{T}$, where $I_{(\phi(\vec{x}), \phi'(\vec{x}))}$ is the collection of geometric formulae $\psi(\vec{x})$ over the signature of $\mathbb{T}$ such that either $(\psi \wedge \phi \vdash_{\vec{x}} \perp)$ or $(\psi \wedge \phi' \vdash_{\vec{x}} \perp)$ is provable in $\mathbb{T}$. On the other hand, in the case of specific theories of natural mathematical interest, it is often possible to establish more natural and 'economical' axiomatizations, by exploiting the specific features of the theory under consideration. For instance, Proposition 2.3 [29] identifies the DeMorganization of the coherent theory of fields as the geometric theory of fields of finite characteristic which are algebraic over their prime fields (and its Booleanization as the theory of fields of finite characteristic which are algebraic over their prime field and

algebraically closed). This surprising result provides by itself a compelling illustration of the ‘centrality’ of topos-theoretic invariants in mathematics, including those which might appear too abstract to be of any ‘concrete’ relevance.

We should mention that there exist natural analogues of the Booleanization or DeMorganization construction for propositional logics other than Boolean and De Morgan logic, including all the usual superintuitionistic ones; these are introduced and characterized in terms of sites in [25].

4.2.4 The dense-closed factorization of a geometric inclusion

Let us continue our analysis of how well-known notions from elementary topos theory transfer to geometric logic through the duality of Theorem 3.2.5.

Recall from section A4.5 of [55] that the dense-closed factorization of a geometric inclusion $\mathbf{sh}_j(\mathcal{E}) \hookrightarrow \mathcal{E}$ is given by

$$\mathbf{sh}_j(\mathcal{E}) \hookrightarrow \mathbf{sh}_{c(\text{ext}(j))}(\mathcal{E}) \hookrightarrow \mathcal{E},$$

where $\text{ext}(j)$ is the c_j -closure of the subobject $0 \rightarrow 1$; the local operator $c(\text{ext}(j))$ is said to be the *closure* of j . In this section we shall interpret the meaning of this construction at the level of sites and then, via the duality of Theorem 3.2.5, in terms of theories.

Let $\mathbf{Sh}(\mathcal{C}, J)$ be a Grothendieck topos, $a_J : [\mathcal{C}^{\text{op}}, \mathbf{Set}] \rightarrow \mathbf{Sh}(\mathcal{C}, J)$ the associated sheaf functor and J' a Grothendieck topology on $\mathcal{C}$ which contains J . Let us calculate the dense-closed factorization of the canonical geometric inclusion $\mathbf{Sh}(\mathcal{C}, J') \hookrightarrow \mathbf{Sh}(\mathcal{C}, J)$ in terms of the corresponding local operator $\tau_{J'}$ on $\mathbf{Sh}(\mathcal{C}, J)$.

The monomorphism $0 \rightarrow 1$ in $\mathbf{Sh}(\mathcal{C}, J)$ is the image of the morphism $0 \rightarrow 1$ in $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$ via the associated sheaf functor $a_J : [\mathcal{C}^{\text{op}}, \mathbf{Set}] \rightarrow \mathbf{Sh}(\mathcal{C}, J)$; from Proposition 4.2.4 and Theorem 4.2.3(iii) it thus follows that the closure of $0 \rightarrow 1$ in $\mathbf{Sh}(\mathcal{C}, J)$ with respect to the local operator corresponding to the geometric inclusion $\mathbf{Sh}(\mathcal{C}, J') \hookrightarrow \mathbf{Sh}(\mathcal{C}, J)$ is equal to the J' -closure of $0 \rightarrow 1$ in $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$. Now, recall from [63] (formula (6) at p. 235) that, for any Grothendieck topology K on $\mathcal{C}$, the K -closure $c_K(A')$ of a subobject $A' \rightarrow E$ in $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$ is given by

$$e \in c_K(A')(c) \text{ if and only if } \{f : d \rightarrow c \mid E(f)(e) \in A'(d)\} \in K(c).$$

Given a subterminal object U of $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$, identified with the full subcategory $\mathcal{C}_U$ of $\mathcal{C}$ as in section 4.2.2.1 above, it is easy to verify that the K -closure $c_K(U \rightarrow 1)$ of $U \rightarrow 1$ in $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$ identifies with the full subcategory $\mathcal{C}_U^K$ on the objects $c \in \mathcal{C}$ such that $\{f : d \rightarrow c \mid d \in \mathcal{C}_U\} \in K(c)$; in particular, if $U = 0$ then the objects of $\mathcal{C}_0^K$ are exactly the objects $c \in \mathcal{C}$ such that $\emptyset \in K(c)$.

By applying this to our topology J' we obtain that $\text{ext}(\tau_{J'})$ identifies (as a subterminal object of $\mathbf{Sh}(\mathcal{C}, J)$) with the J -ideal $\mathcal{C}_0^{J'} = \{c \in \mathcal{C} \mid \emptyset \in J'(c)\}$. Recalling the description of closed local operators on Grothendieck toposes given in section 4.2.2.2, we deduce that the dense-closed factorization of the inclusion $\mathbf{Sh}(\mathcal{C}, J') \hookrightarrow \mathbf{Sh}(\mathcal{C}, J)$ is given by

$$\mathbf{Sh}(\mathcal{C}, J') \hookrightarrow \mathbf{Sh}(\mathcal{C}, J_{\mathcal{C}_{J'}}^{\text{closed}}) \hookrightarrow \mathbf{Sh}(\mathcal{C}, J),$$

where the topology $J_{\mathcal{C}_{J'}}^{\text{closed}}$ is defined by

$$R \in J_{\mathcal{C}_{J'}}^{\text{closed}}(c) \text{ if and only if } Z(c) \cup R \in J(c)$$

(where, for any $c \in \mathcal{C}$, $Z(c) = \{f : d \rightarrow c \mid \emptyset \in J'(d)\}$).

Finally, we study the effect of the dense-closed factorization on theories via the duality of Theorem 3.2.5.

Given a geometric theory $\mathbb{T}$ over a signature Σ and a quotient $\mathbb{T}'$ of $\mathbb{T}$, we can describe the quotient $\mathbb{T}'^{\text{dc}}$ of $\mathbb{T}$ such that

$$\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}'}^{\mathbb{T}}) \hookrightarrow \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}'^{\text{dc}}}^{\mathbb{T}}) \hookrightarrow \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$$

is the dense-closed factorization of the canonical geometric inclusion

$$\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}'}^{\mathbb{T}}) \hookrightarrow \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}}).$$

By equivalence (i) in Remark 3.2.8 we have that $\emptyset \in J_{\mathbb{T}'}^{\mathbb{T}}(\{\vec{y}. \psi\})$ if and only if $(\psi \vdash_{\vec{y}} \perp)$ is provable in $\mathbb{T}'$. So the geometric sentence ϕ corresponding to the subterminal object given by $\mathcal{C}_0^{J_{\mathbb{T}'}}_{\mathbb{T}'}$ is characterized by the property that $\{\square. \phi\} \mapsto \{\square. \top\}$ is the union in $\mathcal{C}_{\mathbb{T}}$ of the images of all the arrows $\{\vec{y}. \psi\} \rightarrow \{\square. \top\}$ such that $(\psi \vdash_{\vec{y}} \perp)$ is provable in $\mathbb{T}'$ (cf. section 4.2.2.1). In view of the results of section 4.2.2.2, we can thus conclude that $\mathbb{T}'^{\text{dc}}$ is obtained from $\mathbb{T}$ by adding the sequent

$$(\phi \vdash_{\square} \perp)$$

or equivalently all the sequents of the form $(\psi \vdash_{\vec{y}} \perp)$ which are provable in $\mathbb{T}'$.

4.2.5 Skeletal inclusions

Recall from section D4.6 of [55] that a geometric morphism $f : \mathcal{F} \rightarrow \mathcal{E}$ is said to be *skeletal* if it restricts to a geometric morphism $\mathbf{sh}_{\neg\neg}(\mathcal{F}) \rightarrow \mathbf{sh}_{\neg\neg}(\mathcal{E})$. By Lemma D4.6.10 [55], a geometric inclusion $f : \mathcal{F} \rightarrow \mathcal{E}$ corresponding to a local operator j on $\mathcal{E}$ is skeletal if and only if $\text{ext}(j)$ (cf. section 4.2.4 for its definition) is a $\neg\neg$ -closed subterminal object of $\mathcal{E}$.

Let us use the notation of section 4.2.4. Given the canonical geometric inclusion $\mathbf{Sh}(\mathcal{C}, J') \hookrightarrow \mathbf{Sh}(\mathcal{C}, J)$ corresponding to an inclusion $J \subseteq J'$, $\text{ext}(\tau_{J'}^J)$ identifies (as a subterminal object of $\mathbf{Sh}(\mathcal{C}, J)$) with the J -ideal $\mathcal{C}_0^J = \{c \in \mathcal{C} \mid \emptyset \in J'(c)\}$. Now, consider the full subcategory $\tilde{\mathcal{C}}$ of $\mathcal{C}$ on the objects which are not J -covered by the empty sieve; $\tilde{\mathcal{C}}$ is J -dense in $\mathcal{C}$ and hence, by the Comparison Lemma (cf. Theorem 1.1.8), $\mathbf{Sh}(\mathcal{C}, J) \simeq \mathbf{Sh}(\tilde{\mathcal{C}}, J|_{\tilde{\mathcal{C}}})$, where $J|_{\tilde{\mathcal{C}}}$ is the induced Grothendieck topology on $\tilde{\mathcal{C}}$. Moreover, $J|_{\tilde{\mathcal{C}}}$ is dense, i.e. $J|_{\tilde{\mathcal{C}}} \leq \neg\neg_{[\tilde{\mathcal{C}}^{\text{op}}, \mathbf{Set}]}$. Therefore, by Corollary 4.2.9 and Theorem 4.2.3(ii), $\text{ext}(\tau_{J'}^J)$ is $\neg\neg_{\mathbf{Sh}(\tilde{\mathcal{C}}, J|_{\tilde{\mathcal{C}}})}$ -closed (as a subterminal object of $\mathbf{Sh}(\tilde{\mathcal{C}}, J|_{\tilde{\mathcal{C}}})$) if and only if $\text{ext}(\tau_{J'}^J)$ is $\neg\neg_{[\tilde{\mathcal{C}}^{\text{op}}, \mathbf{Set}]}$ -closed

(as a subterminal object of $[\tilde{\mathcal{C}}^{\text{op}}, \mathbf{Set}]$). But $\neg\neg_{[\tilde{\mathcal{C}}^{\text{op}}, \mathbf{Set}]}$ is well known to correspond to the dense topology on $\tilde{\mathcal{C}}$ (cf. Example 1.1.4(b)); so, by formula (6) at p. 235 of [63], we obtain that $\text{ext}(\tau_{J'}^J)$ is $\neg\neg_{[\tilde{\mathcal{C}}^{\text{op}}, \mathbf{Set}]}$ -closed if and only if, for any $c \in \tilde{\mathcal{C}}$, $c \in \mathcal{C}_0^{J'}$ if $\{f : d \rightarrow c \text{ in } \tilde{\mathcal{C}} \mid d \in \mathcal{C}_0^{J'}\}$ is stably non-empty in $\tilde{\mathcal{C}}$.

Hence the geometric inclusion $\mathbf{Sh}(\mathcal{C}, J') \hookrightarrow \mathbf{Sh}(\mathcal{C}, J)$ is skeletal if and only if, for any $c \in \tilde{\mathcal{C}}$, $\emptyset \in J'(c)$ if $Z(c) = \{f : d \rightarrow c \text{ in } \tilde{\mathcal{C}} \mid \emptyset \in J'(d)\}$ is stably non-empty in $\tilde{\mathcal{C}}$.

Let us interpret the notion of skeletal inclusion at the level of theories, via the duality of Theorem 3.2.5. Given a geometric theory $\mathbb{T}$ over a signature Σ , we want to characterize the quotients $\mathbb{T}'$ of $\mathbb{T}$ such that the canonical geometric inclusion $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}}^{\mathbb{T}}) \hookrightarrow \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$ is skeletal.

We have that $\emptyset \in J_{\mathbb{T}'}^{\mathbb{T}}(\{\vec{y}. \psi\})$ if and only if $(\psi \vdash_{\vec{y}} \perp)$ is provable in $\mathbb{T}'$. Given an object $\{\vec{y}. \psi\} \in \mathcal{C}_{\mathbb{T}}$, let us denote by $\{\vec{y}. \psi_{\mathbb{T}'}\} \rightarrow \{\vec{y}. \psi\}$ the subobject in $\mathcal{C}_{\mathbb{T}}$ given by the union in $\mathcal{C}_{\mathbb{T}}$ of all the subobjects $\{\vec{y}. \psi'\} \rightarrow \{\vec{y}. \psi\}$ such that $(\psi' \vdash_{\vec{y}} \perp)$ is provable in $\mathbb{T}'$. We thus obtain the following necessary and sufficient condition for $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}}^{\mathbb{T}}) \hookrightarrow \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$ to be skeletal (below by a $\mathbb{T}$ -consistent geometric formula-in-context we mean a geometric formula $\phi(\vec{x})$ such that $(\phi \vdash_{\vec{x}} \perp)$ is not provable in $\mathbb{T}$): for any $\mathbb{T}$ -consistent geometric formula $\psi(\vec{y})$ over Σ , if $(\chi \wedge \psi_{\mathbb{T}'})(\vec{y})$ is $\mathbb{T}$ -consistent for any $\mathbb{T}$ -consistent geometric formula $\chi(\vec{y})$ over Σ such that $(\chi \vdash_{\vec{y}} \psi)$ is provable in $\mathbb{T}$ then $(\psi \vdash_{\vec{y}} \perp)$ is provable in $\mathbb{T}'$.

4.2.6 The surjection-inclusion factorization

Recall that every geometric morphism can be factored, uniquely up to canonical equivalence, as a surjection followed by an inclusion (cf. for instance Theorem A4.2.10 [55]). In this section we shall discuss the meaning of this factorization in terms of theories via the duality of Theorem 3.2.5.

Definition 4.2.12 Let M be a Σ -structure in a Grothendieck topos $\mathcal{E}$. The *theory* $\text{Th}(M)$ of M is the geometric theory over Σ consisting of all the geometric sequents σ over Σ which hold in M .

Theorem 4.2.13 Let $\mathbb{T}$ be a geometric theory over a signature Σ and $f : \mathcal{F} \rightarrow \mathbf{Set}[\mathbb{T}]$ a geometric morphism corresponding to a $\mathbb{T}$ -model M in $\mathcal{F}$ via the universal property of the classifying topos of $\mathbb{T}$. Then the topos $\mathcal{E}$ in the surjection-inclusion factorization $\mathcal{F} \twoheadrightarrow \mathcal{E} \hookrightarrow \mathbf{Set}[\mathbb{T}]$ of f classifies the quotient $\text{Th}(M)$ of $\mathbb{T}$ via Theorem 3.2.5.

Proof Let us denote by $\mathcal{F} \xrightarrow{f'} \mathcal{E} \xrightarrow{i} \mathbf{Set}[\mathbb{T}]$ the surjection-inclusion factorization of f . Since i is a geometric inclusion to the classifying topos of $\mathbb{T}$, i corresponds via the duality of Theorem 3.2.5 to a unique quotient $\mathbb{T}'$ of $\mathbb{T}$ (up to syntactic equivalence) whose classifying topos is $\mathcal{E}$. We want to prove that $\mathbb{T}' = \text{Th}(M)$. From the proof of Theorem 3.2.5 we know that, if U is a universal model of $\mathbb{T}$ in $\mathbf{Set}[\mathbb{T}]$ then $i^*(U)$ is a universal model V for $\mathbb{T}'$. So f' corresponds to the $\mathbb{T}'$ -model M via the universal property of the classifying topos of $\mathbb{T}'$, since $f'^*(V) =$

$f'^*(i^*(U)) \cong f^*(U) = M$. Now, since f' is a surjection, M is a conservative $\mathbb{T}'$ -model (cf. Lemma 1.3.23 and Remark 2.1.9), from which it follows that $\mathbb{T}' = \text{Th}(M)$. $\square$

Remark 4.2.14 The theorem implies that if $\mathbb{T}$ is a geometric theory over a (possibly many-sorted) signature Σ and M is a $\mathbb{T}$ -model in a Grothendieck topos then f_M is a surjection if and only if M is conservative. This result generalizes Corollary D3.2.6 [55], which was proved under the assumption that Σ be one-sorted.

4.2.7 Atoms

In this section we describe the atoms of the lattice of subtoposes of a given elementary topos, that is the non-trivial toposes having no proper subtoposes. Recall that a topos $\mathcal{E}$ is said to be *trivial* if it is equivalent to the category having just one object and the identity morphism on it, equivalently if it is *degenerate* i.e. $0_{\mathcal{E}} \cong 1_{\mathcal{E}}$.

Definition 4.2.15 A topos $\mathcal{E}$ is said to be *two-valued* if it has exactly two (non-isomorphic) subobjects of $1_{\mathcal{E}}$ (namely, the zero subobject and the identity one).

Proposition 4.2.16 *Let $\mathcal{E}$ be an elementary topos. Then the atoms of the lattice of subtoposes of $\mathcal{E}$ are exactly the two-valued Boolean subtoposes of $\mathcal{E}$.*

Proof Our thesis follows as an immediate consequence of the following two facts: first, every non-trivial topos contains a non-trivial Boolean subtopos, and second, a non-trivial Boolean topos does not contain any proper subtoposes if and only if it is two-valued. To prove the first assertion, we observe that if $\mathcal{E}$ is non-trivial then $\mathbf{sh}_{\neg\neg}(\mathcal{E})$ is again non-trivial; indeed, $1_{\mathcal{E}}$ clearly belongs to $\mathbf{sh}_{\neg\neg}(\mathcal{E})$ while $0_{\mathcal{E}}$ belongs to $\mathbf{sh}_{\neg\neg}(\mathcal{E})$ since $\neg\neg$ is a dense local operator on $\mathcal{E}$ (cf. p. 219 of [55]), so if $\mathbf{sh}_{\neg\neg}(\mathcal{E})$ is trivial then $0_{\mathcal{E}} \cong 1_{\mathcal{E}}$ i.e. $\mathcal{E}$ is trivial. The fact that $\mathbf{sh}_{\neg\neg}(\mathcal{E})$ is Boolean is well-known (see for example Lemma A4.5.22 [55]). This completes the proof of the first fact. It remains to prove the second assertion. Given two subterminal objects U and V of $\mathcal{E}$, the subtopos $\mathcal{E}/U \hookrightarrow \mathcal{E}$ is contained in the subtopos $\mathcal{E}/V \hookrightarrow \mathcal{E}$ if and only if $U \leq V$ in the lattice $\text{Sub}_{\mathcal{E}}(1_{\mathcal{E}})$. Indeed, $\mathcal{E}/U \hookrightarrow \mathcal{E}$ factors through $\mathcal{E}/V \hookrightarrow \mathcal{E}$ if and only if the projection $U \times V \rightarrow U$ is isomorphic to the terminal object $1_U : U \rightarrow U$ in $\mathcal{E}/U$ and, since any object admits at most one morphism to a given subterminal object, this condition is equivalent to requiring that $U \leq V$ (equivalently, $U \leq U \times V$). Now, if $\mathcal{E}$ is Boolean then all the subtoposes of $\mathcal{E}$ are open (by Proposition A4.5.22 [55]), whence we have a lattice isomorphism between $\text{Sub}_{\mathcal{E}}(1_{\mathcal{E}})$ and the lattice of subtoposes of $\mathcal{E}$; therefore a non-trivial Boolean topos does not contain any proper subtoposes if and only if it is two-valued. $\square$

Remark 4.2.17 For any Grothendieck topos $\mathcal{E}$ with enough points, $\mathcal{E}$ is Boolean and two-valued if and only if it is atomic and connected. Indeed, we know from

Corollary D3.5.2 [55] that every Boolean Grothendieck topos with enough points is atomic; furthermore, any atomic topos is Boolean and it is two-valued if and only if it is connected (cf. the proof of Theorem 2.5 [23]).

Now we want to understand, via Theorem 3.2.5, the meaning of Proposition 4.2.16 in terms of theories.

Definition 4.2.18 (cf. section 4.1 of [13]) A geometric theory $\mathbb{T}$ is said to be *Boolean* if its classifying topos is a Boolean topos.

Remarks 4.2.19 (a) It is shown in [13] (Theorem 4.2 therein) that a geometric theory $\mathbb{T}$ over a signature Σ is Boolean if and only if for every geometric formula $\phi(\vec{x})$ over Σ there exists a geometric formula $\psi(\vec{x})$ over Σ in the same context such that $\phi \wedge \psi \dashv\vdash_{\mathbb{T}} \perp$ and $\phi \vee \psi \dashv\vdash_{\mathbb{T}} \top$ (where the notation is that of Definition 1.4.1(f)). From this criterion it easily follows that if $\mathbb{T}$ is a Boolean theory then every infinitary first-order formula over Σ is $\mathbb{T}$ -provably equivalent using classical logic to a geometric formula in the same context; indeed, this can be proved by an inductive argument as in the proof of Theorem D3.4.6 [55] (in the case of an infinitary conjunction $\bigwedge_{i \in I} \phi_i$, we observe that this formula is equivalent in classical logic to the

formula $\neg(\bigvee_{i \in I} \neg \phi_i)$, where the symbol $\neg$ denotes the first-order negation).

Notice that, since every infinitary first-order formula is classically equivalent in $\mathbb{T}$ to a geometric formula, it follows from the axioms of infinitary first-order logic for implication and infinitary conjunction that the first-order implication between geometric formulae is classically provably equivalent in $\mathbb{T}$ to the Heyting implication between them in the relevant subobject lattice of $\mathcal{C}_{\mathbb{T}}$; moreover, the infinitary conjunction of a family of geometric formulae is classically provably equivalent in $\mathbb{T}$ to its infimum in that lattice.

- (b) Recall the following fact about elementary toposes (cf. Proposition A4.5.22 [55]): an elementary topos is Boolean if and only if every subtopos of it is open. It is interesting to interpret the ‘only if’ part of this statement at the level of theories via the duality of Theorem 3.2.5. If $\mathbb{T}$ is a Boolean geometric theory over a signature Σ and $\mathbb{T}'$ is a quotient of $\mathbb{T}$, we want to show that there exists a geometric sentence ϕ over Σ such that $\mathbb{T}'$ is syntactically equivalent to $\mathbb{T}_{\phi}$. For any axiom $\sigma \equiv (\phi \vdash_{\vec{x}} \psi)$ of $\mathbb{T}'$, consider the geometric formula $U(\sigma)$ over Σ classically equivalent in $\mathbb{T}$ (as in Remark 4.2.19(a)) to the infinitary first-order sentence $\forall \vec{x}(\phi \Rightarrow \psi)$. Now, there is only a set of such formulae $U(\sigma)$ over Σ up to $\mathbb{T}$ -provable equivalence, the geometric syntactic category $\mathcal{C}_{\mathbb{T}}$ being well-powered, so we can take ϕ to be a geometric sentence which is classically equivalent in $\mathbb{T}$ to their infinitary conjunction; it is immediate to see that ϕ has the required property.

Proposition 4.2.20 (Proposition 3.3 [18]). *Let $\mathbb{T}$ be a geometric theory over a signature Σ . Then $\mathbb{T}$ has enough set-based models (in the sense of Definition 1.4.17) if and only if its classifying topos $\mathbf{Set}[\mathbb{T}]$ has enough points.*

Proof By definition, $\mathbf{Set}[\mathbb{T}]$ has enough points if and only if the inverse image functors f^* of the geometric morphisms $f : \mathbf{Set} \rightarrow \mathbf{Set}[\mathbb{T}]$ are jointly conservative.

Since the geometric morphism $f_M : \mathbf{Set} \rightarrow \mathbf{Set}[\mathbb{T}]$ corresponding to a $\mathbb{T}$ -model M in $\mathbf{Set}$ satisfies $f^*(U_{\mathbb{T}}) = M$ (where $U_{\mathbb{T}}$ is ‘the’ universal model of $\mathbb{T}$ lying in $\mathbf{Set}[\mathbb{T}]$), if $\mathbf{Set}[\mathbb{T}]$ has enough points then any geometric sequent over Σ which is satisfied in every $\mathbb{T}$ -model M in $\mathbf{Set}$ is satisfied in $U_{\mathbb{T}}$, equivalently it is provable in $\mathbb{T}$ (cf. Remark 2.1.9).

Conversely, if $\mathbb{T}$ has enough set-based models then, since the objects of the geometric syntactic category $\mathcal{C}_{\mathbb{T}}$ form a separating family of objects for the topos $\mathbf{Set}[\mathbb{T}] \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$, the inverse image functors of the points of $\mathbf{Set}[\mathbb{T}]$ jointly reflect isomorphisms of subobjects (equivalently, $\mathbf{Set}[\mathbb{T}]$ has enough points). $\square$

Definition 4.2.21 A geometric theory $\mathbb{T}$ over a signature Σ is said to be *complete* if every geometric sentence ϕ over Σ is $\mathbb{T}$ -provably equivalent to $\top$ or $\perp$, but not both.

Remark 4.2.22 A geometric theory $\mathbb{T}$ over a signature Σ is complete if and only if its classifying topos is two-valued; indeed, as we observed in section 4.2.2.1, the subterminal objects of the classifying topos $\mathbf{Set}[\mathbb{T}]$ can be identified with the $\mathbb{T}$ -provable equivalence classes of geometric sentences over Σ .

Atomic toposes were originally introduced in [4]. They can be characterized as the Grothendieck toposes whose objects can be written as coproducts of *atoms*, i.e. of non-zero objects which do not have any proper subobjects, or equivalently as the toposes which can be represented as categories of sheaves on a site of the form $(\mathcal{C}, J_{\text{at}})$, where $\mathcal{C}$ is a category satisfying the right Ore condition and J_{at} is the atomic topology on it. Every atomic topos is Boolean and every Boolean topos with enough points is atomic (cf. Corollary C3.5.2 [55]).

Definition 4.2.23 A geometric theory $\mathbb{T}$ is said to be *atomic* if its classifying topos $\mathbf{Set}[\mathbb{T}]$ is an atomic topos.

Recall that a geometric theory $\mathbb{T}$ over a signature Σ is said to be *contradictory* if the sequent $(\top \vdash_{\square} \perp)$ is provable in $\mathbb{T}$.

Remark 4.2.24 A geometric theory is contradictory if and only if its classifying topos is trivial. Indeed, if $\mathbb{T}$ is contradictory then the trivial topos satisfies the universal property of the classifying topos of $\mathbb{T}$; conversely, if the classifying topos of $\mathbb{T}$ is trivial then $\perp$ holds in it and hence $(\top \vdash_{\square} \perp)$ is provable in $\mathbb{T}$.

The following proposition represents the translation of Proposition 4.2.16 in terms of theories via Theorem 3.2.5.

Proposition 4.2.25 *Let $\mathbb{T}$ be a geometric theory over a signature Σ . Then the maximal quotients of $\mathbb{T}$ (i.e. the non-contradictory quotients $\mathbb{T}'$ of $\mathbb{T}$ such that for every geometric sequent σ over Σ either σ is provable in $\mathbb{T}$ or the theory $\mathbb{T} \cup \{\sigma\}$ is contradictory) are exactly the Boolean and complete ones.* $\square$

- Remarks 4.2.26** (a) The ‘if’ direction in Proposition 4.2.25 can be easily proved without appealing to Theorem 3.2.5, as follows. If $\mathbb{T}$ is Boolean then a geometric sequent $(\phi \vdash_{\bar{x}} \psi)$ over Σ is provable in $\mathbb{T}$ if and only if the infinitary first-order sentence $\forall \bar{x}(\phi \Rightarrow \psi)$ is. But by Remark 4.2.19 this formula is $\mathbb{T}$ -provably equivalent using classical logic to a geometric sentence, and this sentence is $\mathbb{T}$ -provably equivalent to $\top$ or $\perp$ since $\mathbb{T}$ is complete.
- (b) The theories whose classifying toposes admit the Galois-type representations described in section 10.5 satisfy the equivalent conditions of the proposition (when considered as quotients of the empty theory over their signature); indeed, their classifying toposes are atomic and two-valued. More generally, the maximal quotients of an atomic theory $\mathbb{T}$ are precisely the theories of the form $\mathbb{T} \cup \{(\top \vdash_{\square} \phi)\}$, where ϕ is any of the geometric formulae arising in the decomposition of the terminal object $\{\square, \top\}$ of $\mathbf{Set}[\mathbb{T}]$ as a coproduct of atoms (cf. also Remark 10.4.6).

4.2.8 Subtoposes with enough points

We have seen that the classifying topos $\mathbf{Set}[\mathbb{T}]$ of a geometric theory $\mathbb{T}$ has enough points if and only if $\mathbb{T}$ has enough set-based models (cf. Proposition 4.2.20).

Given a point p of a topos $\mathcal{E}$, let us denote by $\mathcal{E}_p \hookrightarrow \mathcal{E}$ the inclusion part of the surjection-inclusion factorization of p . By Theorem 4.2.13, if $\mathcal{E} = \mathbf{Set}[\mathbb{T}]$ then $\mathcal{E}_p$ classifies $\mathrm{Th}(M)$ where M is the $\mathbb{T}$ -model corresponding to p .

Given a Grothendieck topos $\mathcal{E}$, let us define the *subtopos $\mathcal{E}^{\mathrm{points}}$ of points* of $\mathcal{E}$ to be the union of all the subtoposes $\mathcal{E}_p$ of $\mathcal{E}$ as p varies among the points of $\mathcal{E}$ (such a union exists because, dually, any intersection of Grothendieck topologies is a Grothendieck topology).

From Theorem 3.2.5 and the description of the (infinitary) meet in $\mathfrak{H}_{\Sigma}^{\mathbb{T}}$, it follows that the topos $\mathbf{Set}[\mathbb{T}]^{\mathrm{points}}$ classifies the intersection of all the theories $\mathrm{Th}(M)$ as M varies among the $\mathbb{T}$ -models M in $\mathbf{Set}$; in particular $\mathbf{Set}[\mathbb{T}]$ coincides with $\mathbf{Set}[\mathbb{T}]^{\mathrm{points}}$ if and only if it has enough points. Notice that, obviously, any intersection in $\mathfrak{H}_{\Sigma}^{\mathbb{T}}$ of theories of the form $\mathrm{Th}(M)$ (for a $\mathbb{T}$ -model M in $\mathbf{Set}$) has enough set-based models; so all the toposes of the form $\mathbf{Set}[\mathbb{T}]^{\mathrm{points}}$ have enough points. Therefore, given a geometric theory $\mathbb{T}$, the quotients of $\mathbb{T}$ having enough set-based models are exactly the intersections in $\mathfrak{H}_{\Sigma}^{\mathbb{T}}$ of theories of the form $\mathrm{Th}(M)$ (where M is a $\mathbb{T}$ -model in $\mathbf{Set}$). In light of the fact that every Grothendieck topos is (equivalent to) the classifying topos of a geometric theory, this translates into the following equivalent topos-theoretic statement: the subtoposes of a Grothendieck topos $\mathcal{E}$ which have enough points are exactly the unions of subtoposes of the form $\mathcal{E}_p$ where p is a point of $\mathcal{E}$.

Finally, we note that, given an atom $\mathcal{F}$ in the lattice of subtoposes of a Grothendieck topos $\mathcal{E}$, i.e. a Boolean and two-valued subtopos $\mathcal{F}$ of $\mathcal{E}$ (cf. section 4.2.7 above), if $\mathcal{F}$ has enough points then $\mathcal{F}$ is of the form $\mathcal{E}_p$ for a point p of $\mathcal{E}$. Indeed, a topos with enough points has a point if and only if it is non-trivial.